

IDEA League

Master of Science in Applied Geophysics
Research Thesis

From Particles to Waves

**Towards coupling Dynamic Snow Modelling and Spectral Elements to
detect Avalanche Breakoff in DAS Data.**

Salome Anna Bachmann

Wednesday July 29, 2026

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Master of Science Thesis

for the degree of Master of Science in Applied Geophysics
by

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IDEA LEAGUE
JOINT MASTER'S IN APPLIED GEOPHYSICS

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Abstract

Avalanches are the most fatal natural hazard in Switzerland (LongtermStatistics), and while processes like crack propagation, the transition between crack propagation regimes and the influence of snow and terrain parameters have been studied, there is no link between crack propagation and its seismic signature. In this master thesis, spectral element modelling in SALVUS will be used to simulate crack propagation in snow, as measured by distributed acoustic sensing (DAS). The spectral element modelling results will encompass simple simulations of sub-rayleigh and supershear propagation of uniform source wavelets in snow to determine the different types of waves in snow. Furthermore, the results from material point method simulations will be used to obtain seismic moment tensors and source wavelet shapes in an attempt to make the crack propagation model more physical.

The simple model results show that the main energy in the results is focused at the propagation speed of the crack. In the simple sub-rayleigh case, there is a precursor to the main energy front, which has been identified to be the S0 wave. This precursor is caused by lack of interference of the first source at dense spacing. Overall, sub-rayleigh regimes have lower energy than supershear propagation regimes, which is important because they might not be detected in DAS, especially if there is a transition in propagation speed, as observed by trotterTransitionSubRayleighAnticrack2022. Another difference between supershear and sub-rayleigh crack propagation is that supershear wavefronts decay in amplitude with time, whereas sub-rayleigh have a constant amplitude. Both regimes show a rupture arrest front, which occurs in different modes. Supershear cracks have a fan-like front which only extends in the positive direction, whereas sub-rayleigh has a v-shaped arrest front which propagates in both the positive and negative direction. This is important for identifying the propagation regime in DAS field data.

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Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (schweizerSnowAvalancheFormation2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities in Switzerland, which are caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (LongtermStatistics). These data show that avalanches are a significant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (anceySnowAvalanches2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (schweizerSnowAvalancheFormation2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (schweizerSnowAvalancheFormation2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (gaumeSnowFractureRelation2017a). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (fohnMechanicalStructuralProperties1998).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain for potential avalanches.

Furthermore, detecting slab breakoff with distributed acoustic sensing is also important to further understand crack dynamics in snow as well as the influences on crack propagation, such as snowpack properties.

In this master's thesis, the aim is to create an accurate spectral element model (SE) for crack propagation in snow. This is achieved through two approaches: pure spectral element (SE) simulations, which have simple source-time functions in order to set up the model and to understand the underlying physics. These are then combined with simulations which are driven by moment tensor STFS which are extracted from MPM simulations. This helps towards understanding the processes associated with crack propagation in snow, that eventually lead to slab breakoff.

In this thesis, a fibre-optic cable installed on a slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (parkerDistributedAcousticSensing2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (shatalinInterferometricOpticalTimedomain2017). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (hartogIntroductionDistributedOptical2017). The phase variations are converted into dynamic strain along the cable (hartogIntroductionDistributedOptical2017). Strain is the first spatial derivative of deformation, so deformation is the integral in respect to space of strain (trabattoniStrainDisplacementUsing2023).

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (turquetAutomatedSnowAvalanche2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the Swiss Alps (paitzPhenomenologyAvalancheRecordings2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by paitzPhenomenologyAvalancheRecordings2023 and edmeFiberopticDetectionSnow2023, but the detectability of the slab breakoff itself has not been studied. The same is true for the seismic signatures associated with crack propagation leading to slab breakoff. This is important because the time between initial crack propagation and slab breakoff is crucial for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

A multitude of models have been previously used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (brunNumericalModelSimulate1992; barteltPhysicalSNOWPACKModel2002). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient or water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (brunNumericalModelSimulate1992; barteltPhysicalSNOWPACKModel2002). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and the consequences they have for the structure of snow. While it is important to understand the influences of for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Furthermore, SNOWPACK and CROCUS do not model the seismic signal itself, which is what is needed to detect and understand crack propagation in DAS. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. cundallDiscreteNumericalModel1979 developed the discrete element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (cundallDiscreteNumericalModel1979).

The interactions of particles are modeled by the governing equations and the resulting forces and displacements (cundallDiscreteNumericalModel1979). This method has been used by bobillierMicromechanicalModelin bobillierNumericalInvestigationCrack2024; mulakNumericalInvestigationMixedmode2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (cundallDiscreteNumericalModel1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (gaumeInvestigatingReleaseFlow2019).

The same is true for finite element modelling (FE or FEM), which is done using SALVUS in this Thesis. FEM has been used to model wave propagation in snow since the 1970s, a review of the different approaches has been published by podolskiyReviewFiniteelementModelling2013. It is important to note that the FE approach used in this thesis is concerned with wave propagation in snow and not the mechanics of snow itself, so it is fundamentally different from other methods. The finite element modelling approach used in this thesis is described more in detail in ???. FE can also not account for anticrack propagation, similarly to DEM, mesh elements would need to be removed to account for anticrack propagation.

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (bardenhagenMaterialpointMethodGranular2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (bardenhagenMaterialpointMethodGranular2000). This method is not only more computationally efficient than DEM on the slope-scale, it can also account for anticrack behavior and strain softening of snow (gaumeInvestigatingReleaseFlow2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by bergfeldSignaturesSubRayleighSupershear2025 who modeled slope-scale experiments to understand the transition of crack propagation modes, trotterTransitionSubRayleigh2025 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as gaumeInvestigatingReleaseFlow2019 and gaumeDynamicAnticrackPropagation2018.

While MPM is very good at accounting for anticrack propagation, FE is more computationally efficient and accurate (lianAdaptiveFiniteElement2012). This is relevant especially for large-scale simulations, such as the ones done in this Thesis. This is because in MPM, the discretization through particles causes the displacement of each particle to be tracked and the contact at each particle to be calculated (zhangModellingLargescaleLandslide2023). Furthermore, MPM does not include boundary conditions in the way FE does (coombsImmersedFEMMPM2025). Finite element is more efficient on the slope scale because the mesh is fixed, as described by (FiniteElementMethod). This means that the degrees of freedom and thus the number of calculations which need to be done are much smaller than in the MPM case. This means that MPM is more accurate at representing snow as a medium with its porous character and anticrack behavior, but FE is more efficient at performing large-scale simulations.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (bergfeldSignaturesSubRayleighSupershear2025).

There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (heierliAnticracksNewTheory2003). In anticrack propagation, the weak layer consolidates under loading (knightSuperheatedIceTrue1972), while cracks propagate in a combination of modes I-III (heierliAnticracksNewTheory2003). gaumeDynamicAnticrackPropagation2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (gaumeDynamicAnticrackPropagation2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models such as mulakNumericalInvestigationMixedmode2019; gaumeInvestigatingReleaseFlow2019 also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (gaumeInvestigatingReleaseFlow2019). Furthermore, there is a directionality in propagation speed, simulations by gaumeInvestigatingReleaseFlow2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (gaumeInvestigatingReleaseFlow2019). Slope angle also influences which mode of crack propagation is more dominant (rosendahlModelingSnowSlab2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (rosendahlModelingSnowSlab2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (rosendahlModelingSnowSlab2020). Snow parameters also have an influence on this critical force, as shown by simulations by rosendahlModelingSnowSlab2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (trottetTransitionSubRayleighAnticrack2022; bobillierNumericalInvestigationCrack2024; bergfeldSignaturesSubRayleighSupershear2025). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, trottetTransitionSubRayleighAnticrack2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (trottetTransitionSubRayleighAnticrack2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (trottetTransitionSubRayleighAnticrack2022). The observations were also made by bobillierNumericalInvestigationCrack2024 as well as bergfeldSignaturesSubRayleighSupershear2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (trottetTransitionSubRayleighAnticrack2022; bergfeldSignaturesSubRayleighSupershear2025). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (gaumeDynamicAnticrackPropagation2018; gaumeSnowFractureRelation2017a; gaumeModelingCrackPropagation2015). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, meaning the lower its Young's Modulus, the slower crack propagation; the

thicker the weak layer, the lower the crack propagation distance before failure (gaumeModelingCrackPropagation2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (gaumeModelingCrackPropagation2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (gaumeSnowFractureRelation2017a). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (gaumeSnowFractureRelation2017a) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (mulakNumericalInvestigationMixedmode2019). This means that new snow contacts can form, which can strengthen the material (mulakNumericalInvestigationMixedmode2019). This is in contrast to rapid loading, which happens for example due to a skier (mulakNumericalInvestigationMixedmode2019).

Because each of the variables described such as weak layer thickness, critical crack length and so on, alter the velocity, geometry, and energy release of the failure in the snowpack, they directly control the characteristics of the resulting seismic signature. In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (afanasievSalvusFlexibleHighperformance2017).

The different snow processes described have different implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seismic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

This thesis will focus on the modelling of crack propagation, as recorded by DAS. Sub-Rayleigh as well as supershear crack propagation will be modelled using different methods and models. The aim is to determine what the different cases of crack propagation will look like in DAS data. This is then relevant for detecting actual avalanches in real-world data. The main questions are what crack propagation in different regimes looks like, how it differs when different modeling methods are used, and how the results can be explained in terms of wave physics.

The main questions investigated in this thesis are:

1. What FE modelling framework is required to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
2. What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-rayleigh versus supershear crack propagation?

3. To what extent do the results change if source wavelets from a MPM simulation of a PST are injected?
4. How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Theory

2.1 Spectral Element Wave Propagation Modeling

The model described in ?? is implemented using the finite-element (FE) method in SALVUS. The following section describes the physics used by SALVUS. Understanding the physics behind the modeling has relevance for understanding and interpreting the results, but also what can be modeled. To discretize a domain, the domain is divided into a finite number of elements (FiniteElementMethod). The number of elements in this case is defined by the maximum wave frequency that needs to be resolved accurately, the number of mesh elements per wavelength is user-input. The boundaries of these elements describe the mesh, the mesh has a finite number of grid-points (edge points of the elements) (FiniteElementMethod). The governing equations are determined at every element node,

so for rectangular elements with nodes 1,2,3 and 4, there is the system $r^1 = \begin{pmatrix} r_1^1 \\ r_2^1 \\ r_3^1 \\ r_4^1 \end{pmatrix}$ for force and

$u^1 = \begin{pmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \\ u_4^1 \end{pmatrix}$ for displacement (FiniteElementMethod). The residual or characteristic relationship of

linear elastic elements is of the form $r^1 = f^1 - K^1 u^1$, where f are the nodal forces distributing load on the nodes and K is the stiffness matrix of a specific element (FiniteElementMethod). In FE, the approximation of the wavefield by basis functions leads to a representation over the whole domain and not just at a set of points (afanasievModularFlexibleSpectralelement2019). This means integrals can be calculated over surfaces or even volumes (afanasievModularFlexibleSpectralelement2019).

Snow is a porous medium (mellorEngineeringPropertiesSnow1977), which is approximated as a solid in this Thesis. This is because SALVUS meshes need an isotropic, solid material and cannot account for pore spaces. Higher frequencies in the resulting data would be attenuated much quicker in an actual porous medium (koushikDevelopmentSeismicSignalbased2024). In addition, the presence of pore spaces in the medium would lead to complex scattering of the fields in the results, their absence is an important factor to consider when comparing simulated data to real-world data. The presence of air or water in snow is not neglected completely in the SALVUS simulations implemented however, because the material properties taken from guillemotEffectSnowfallChanges2021 are based on field measurements of snow. This means that the velocities measured by guillemotEffectSnowfallChanges2021 are not of pure snow, but snow that contains air or water.

The governing equation considered in this case is the elastic wave equation, because real-life crack propagation emits P- and S-waves. The wave equation for a heterogenous elastic solid in the n -dimensional domain G is

$$\rho \partial_t^2 u = \nabla \cdot \sigma + f \quad (2.1)$$

(afanasievModularFlexibleSpectralelement2019; kAkiRichards19801981). Stress can be related to displacement using Hooke's law: $\sigma = C : \nabla u$, where C is a 4th order tensor which characterizes the stiffness of the medium (afanasievModularFlexibleSpectralelement2019). Combining ?? with Hooke's law, this leads to

$$\rho \partial_t^2 u = \nabla \cdot \frac{C}{\nabla u} + f, \quad (2.2)$$

which is the strong form of the linear elastic wave equation.

Boundary conditions, such as free surfaces, as well as initial conditions must be stated explicitly, they have the form $\sigma \cdot \hat{n}|_{\partial G} = \bar{\tau}(x)$, $u|_{t=0} = \bar{u}(x)$, $v|_{t=0} = \bar{v}(x)$ (afanasievModularFlexibleSpectralelement2019). In this case, $\bar{\tau}(x)$, $\bar{u}(x)$, $\bar{v}(x)$ are the spatially variable distributions of traction, displacement and velocity (afanasievModularFlexibleSpectralelement2019). The outward pointing normal vector on the boundary of domain G is described by $\hat{n}|_{\partial G}$.

FE methods use the weak form of the governing equations (FiniteElementMethod). To obtain the weak form of the linear elastic wave equation in ??, the whole equation needs to be multiplied with a test function w , which is not time-dependent (afanasievModularFlexibleSpectralelement2019). The weak form enforces the solution of the governing equation across the domain, instead of solving the equation everywhere, an integration over time leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_G w \cdot (\nabla \cdot \sigma) \, d^n x + \int_G w \cdot f \, d^n x \quad (2.3)$$

(afanasievModularFlexibleSpectralelement2019). The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in an enclosed volume: $\iiint_V (\nabla \cdot F) dV = \oint_{\partial V} (F \cdot \hat{n}) dS$, where F is the vector field. When this is applied to the first term on the right in ??, this leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_{\partial G} w \cdot (\sigma \cdot \hat{n}) \, d^{n-1} x - \int_G (\nabla w : \sigma) \, d^n x + \int_G w \cdot f \, d^n x \quad (2.4)$$

(afanasievModularFlexibleSpectralelement2019). The first integral on the left side of ?? is the boundary integral, which changes depending on the boundary conditions implemented. The integral on the right hand side is the mass term, the second on the left hand side is the stiffness term, and the last on the left hand side is the external forcing term, all in ?? . Equation ?? can be compacted to

$$(\rho \partial_t^2 u, w)_G - \langle \sigma \cdot \hat{n}, w \rangle_{\partial G} + (\sigma, \nabla w)_G = (f, w)_G, \quad (2.5)$$

using the notations $(a, b)_G = \int_G a \cdot b \, d^n x$, $\langle a, b \rangle_{\partial G} = \int_{\partial G} a \cdot b \, d^{n-1} x$ as well as Hooke's law (afanasievModularFlexibleSpectralelement2019). This can be even more compacted by describing a mathematical model I with M for mass term, B for boundary term, S for stiffness and F for forcing term:

$$I_I(M) - I_I(B) + I_I(S) = I_I(F) = \sum_{P \in \{M, B, S, F\}} I_I(P) = 0. \quad (2.6)$$

Physically, this means that the entire model can be described as a combination of the different terms, which is very useful, especially if one of the terms is negligible (afanasievModularFlexibleSpectralelement2019). This means that the same codebase can handle acoustic, elastic and viscoelastic media (afanasievModularFlexibleSpectral because different terms of equation ?? can be set to zero for different materials.

The dependence on the reference coordinate system is described by E , which is a model containing the physical coordinate to reference coordinate transformation, in other words, the determinant of the Jacobian matrix and its inverse (afanasievModularFlexibleSpectraelement2019). Using basis B of the polynomial functions of the approximation V_h , which are piecewise per-element because of possible discontinuities, the abstract form in ?? becomes

$$\sum_{P \in \{M, B, S, F\}} I_I(P, B(E)) = 0 \quad (2.7)$$

(afanasievModularFlexibleSpectraelement2019). This means that fields such as displacement and velocity are calculated in this way. From these fields, the data from the model in section ?? is obtained and plotted.

2.2 Cohesive Zone Physics

The cohesive zone (CZM) model is set up according to mahajanModelingInterfacialCrack2008. This model was set up in lieu of the simpler model to take more complex physics into account. Cohesive tractions are prescribed on zero-thickness interface elements and depend on the normalized opening $\xi = \frac{\delta}{\delta_c}$. The zero-thickness interface elements are implemented by not solving for volumetric strain but displacement between interfaces instead. For the linear softening law used here the traction in a given cohesive element follows

$$T(\xi) = T_{\text{peak}} (1 - \xi) \quad \text{for } 0 \leq \xi \leq 1, \quad T(\xi) = 0 \quad \text{for } \xi > 1,$$

where T_{peak} is the peak cohesive traction and δ_c the critical opening. The normalized opening can be related to the local crack coordinate as $\xi = \frac{(x-x_0)}{l_{\text{crack}}}$ for a crack front with initial position x_0 and local process length l_{crack} .

The mixed-mode potential used in the code follows the form from mahajanModelingInterfacialCrack2008. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the potential is written as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left\{ \left[1 + \frac{\Delta_n}{\delta_n}\right] + q \left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right] \right\},$$

from which the normal and tangential tractions are obtained, the implementation uses the same closed-form expressions. This is so that peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear works of separation relate to peak strengths via

$$\phi_n = e \sigma_{\text{max}} \delta_n, \quad \phi_t = \sqrt{\frac{e}{2}} \tau_{\text{max}} \delta_t,$$

and the critical openings for pure modes are computed as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \quad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

with the mixed-mode critical opening

$$\delta_c = \sqrt{(1 - r_{\text{mode}}) \delta_{n,c}^2 + r_{\text{mode}} \delta_{s,c}^2}.$$

Lamé parameters are derived from input seismic speeds and density: $v_s = \sqrt{\frac{\mu}{\rho}}$ so that $\mu = \rho v_s^2$, and $v_p = \sqrt{\frac{(\lambda+2\mu)}{\rho}}$ so that $\lambda = \rho(v_p^2 - 2v_s^2)$. Young's modulus is then computed from the Lamé

parameters: $E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$ (kAkiRichards19801981). These conversions are applied for every material in the model.

The critical opening distances for pure normal (Mode I) and shear (Mode II) separations are computed from the corresponding fracture energies, strenghts, peak tractions as the mode opening equations and combined for mixed-mode behaviour using

$$\delta_c = \sqrt{(1 - r_{\text{mode}}) \delta_{n,c}^2 + r_{\text{mode}} \delta_{s,c}^2},$$

where $r_{\text{mode}} \in [0, 1]$ controls the mode I/II mix, where 0 is pure mode I and 1 is pure mode II. In the current implementation the slip on each subfault follows the prescribed linear slip law up to the critical slip distance. The mode mix is an user-input parameter, while all the other parameters are derived from laboratory experiments from mahajanModelingInterfacialCrack2008.

The characteristic Irwin length for shear is calculated as $l_{ch} = \frac{E_{\text{eff}} G_{II}}{\tau_s^2}$, and the cohesive zone length is taken as $l_{cz} = \frac{\pi}{8} l_{ch}$, which sets the spatial extent over which cohesive weakening is active ahead of the crack tip.

Rupture propagation speed is chosen with respect to the local shear-wave speed and the bounds discussed in trotterTransitionSubRayleighAnticrack2022. Where appropriate, the rupture speed is corrected for finite slab thickness and bending stiffness to account for slab flexure effects. The rupture front is discretized into a sequence of subfaults: each subfault is activated at an onset time

$$t_o = \frac{x - x_{\text{start}}}{v_{\text{rupture}}},$$

using the subfault horizontal position x , the rupture starting position x_{start} , and the prescribed rupture speed v_{rupture} .

The sources are then set up as described in the methods section. A visual representation of the physics and setup workflow can be seen in appendix ??.

2.3 Wave Propagation Physics

The snow layer in this thesis acts as a thin layer bounded by a free surface above (snow-air interface) and a rigid boundary at the bottom (either domain boundary or bedrock interface). This geometry supports the appearance of Lamb waves: Lamb waves are elastic guided waves that occur in plates (suFundamentalsAnalysisLamb2009) and their amplitude is shaped by layer boundaries (lambWavesElasticPlate1917). They are guided by the boundaries of the layer or media in which they propagate and have a symmetric (S0) and an antisymmetric (A0) mode (suFundamentalsAnalysisLamb2009). The symmetric mode dominates the in-plate motion, while the antisymmetric mode dominates the vertical (out of plane motion) (suFundamentalsAnalysisLamb2009). A sketch of the propagation of these two modes, taken from cartaLambWavesDiscrete2023, can be seen in appendix ??. Both modes are observed in the simulation results shown in ??, and to interpret and understand the results correctly, their properties such as speeds, dispersion relations and excitation need to be understood. This section derives the dispersion relations of both modes and explains the depth-dependent amplitude structure that determines how each mode decays between the crack plane and the snow bottom.

2.3.1 Potentials and wave equations

Starting from the equation of motion for a homogeneous elastic solid used in SALVUS,

$$\mu \nabla^2 u + (\lambda + \mu) \nabla (\nabla \cdot u) = \rho \frac{\partial^2 u}{\partial t^2}, \quad (2.8)$$

the displacement field is decomposed into scalar (ϕ , P-wave) and vector (ψ , S-wave) potentials via the Helmholtz decomposition (achenbachWavePropagationElastic2012):

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial y}, \quad u_y = \frac{\partial \phi}{\partial y} + \frac{\partial \psi}{\partial x}. \quad (2.9)$$

Substituting into equation ?? decouples the problem into separate wave equations for ϕ and ψ , each propagating at the P-wave speed c_p and S-wave speed c_s respectively. Assuming harmonic plane wave solutions of the form $\phi = f(z) e^{i(kx - \omega t)}$ and $\psi = g(z) e^{i(kx - \omega t)}$, the depth functions satisfy

$$\frac{d^2 f}{dz^2} + p^2 f = 0, \quad \frac{d^2 g}{dz^2} + q^2 g = 0, \quad (2.10)$$

with $p^2 = \omega^2/c_p^2 - k^2$ and $q^2 = \omega^2/c_s^2 - k^2$. The general solutions are sinusoidal in z (depth) when $p^2, q^2 > 0$ (propagating through the layer thickness), and hyperbolic when p^2 or $q^2 < 0$ (evanescent decay with depth). The evanescent case applies to wave components whose horizontal phase velocity is slower than the bulk wave speeds, which is the A0 mode at low frequency-thickness product, like it is the case in this thesis and explains why the A0 wave train decays more rapidly between crack depth and snow bottom than the faster S0 mode (graffWaveMotionElastic1975).

2.3.2 Boundary conditions and the Lamb dispersion relations

At the snow-air interface, the large impedance contrast produces a stress-free boundary: normal stress $\sigma_{zz} = 0$ and shear stress $\sigma_{xz} = 0$ at the top and bottom faces of the slab (igelComputationalSeismologyPractical2016). Applying these conditions to the general potentials yields two types of solutions depending on the symmetry of the displacement field across the slab midplane, so across the horizontal axis of the slab.

For the symmetric mode (S0), horizontal displacement is even in z and vertical displacement is odd. Setting the determinant of the resulting 2×2 system to zero gives the S0 dispersion relation:

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2 pq}{(q^2 - k^2)^2}, \quad (2.11)$$

where h is the half-thickness of the slab (suFundamentalsAnalysisLamb2009).

For the antisymmetric mode (A0), the displacement symmetries are reversed: vertical displacement is even in z and horizontal displacement is odd. The A0 dispersion relation is:

$$\frac{\tan(ph)}{\tan(qh)} = \frac{4k^2 pq}{(q^2 - k^2)^2}. \quad (2.12)$$

Both relations give phase velocity as function of frequency, meaning the two modes are dispersive: different frequency components propagate at different speeds (kAkiRichards19801981). At low frequency-thickness product ($f \cdot h \ll 1$, as in this case where $f \cdot h \approx 6$ Hz·m), A0 is strongly dispersive and slow, while S0 is nearly non-dispersive and converges to the plate wave speed with $c_p^{\text{plate}} = 2c_s \sqrt{1 - v_s^2/v_p^2}$ (roseWavesPlates2014). This is why S0 appears as a narrow, fast precursor in the results and A0 produces the broad, dispersive wave train.

2.3.3 Mode symmetry and depth dependence

Because the crack source in the slab middle or weak layer acts as an asymmetric load applied below the slab midplane, it couples energy into both modes simultaneously (roseWavesPlates2014; achenbachWavePropagationElastic2012). The two modes have distinct depth profiles: S0 carries horizontal strain or velocity that is nearly uniform through the slab thickness, while A0 horizontal strain / velocity is proportional to distance from the neutral axis and is therefore maximum at the plate boundaries (graffWaveMotionElastic1975).

2.4 Static and Dynamic Components of Seismic sources

A moment tensor source, such as the sources used in this thesis, has two contributions to the wavefield (kAkiRichards19801981). There is a dynamic component, which propagates away from the source as waves, and the static component (DC), this component represents the permanent deformation field after the sources have stopped firing (freundDynamicFractureMechanics1998; zhuNoteDynamicStatic2002; archambeauGeneralTheoryElastodynamic1968).

The static displacement field of a crack with half length a , in an elastic medium decays with $1/r^2$ with distance r from the crack tip, and produces a permanent strain in the surrounding medium proportional to the total slip δ and the shear modulus μ (freundDynamicFractureMechanics1990):

$$\epsilon_{xx}^{\text{static}} \propto \frac{\mu \delta a}{r^2}. \quad (2.13)$$

This strain accumulates as the crack grows in length, but does not propagate away from the crack location, which means it stays in the near-field (freundDynamicFractureMechanics1998). This occurs because there are no restoring forces, the crack, which has opened, is already at equilibrium.

The control on whether a static field occurs is the shape of the STF. The DC component of the STF, which is its integral over time or the moment rate, has to be non-zero in order for there to be a static component (kAkiRichards19801981; zhuNoteDynamicStatic2002). For a Ricker wavelet source for example, the integral over time is zero because the positive and negative lobes are equal in area and they cancel out. For more irregular shaped source wavelets, as described by kaganPointSourcesElastic1987, like for example the MPM STFs described in the next section, this is not true and the moment rate is non-zero.

2.5 From Particles to Waves: Translating MPM to SE

The material point simulations of the PST performed by (gaumeDynamicAnticrackPropagation2018; trotterTransitionSubRayleighAnticrack2022) result in the recorded stresses in the model according to gaumeInvestigatingReleaseFlow2019. The output of the model is particle velocity, the stress tensor, strain and the J-value, which distinguished between elastic and plastic deformation. The J-value is used as a filter to remove any irreversible deformation such as the weak layer collapse. Only the elastic stress field outside the plastic zone is kept, ensuring that the moment tensor represents the elastic source rather than the inelastic crack process zone itself. This also ensures that any signals emitted from the crack are not taken into account, because inputting them in SALVUS would mean that waves emitted by the crack are used as sources.

To translate the continuum source to something which can be input to SALVUS, the relationship between stress and strain must be defined. The deformation of the material is described by the infinitesimal strain tensor ϵ_{ij} , which is linearly related to the Cauchy stress tensor σ_{ij} via the fourth-order stiffness tensor C_{ijkl} (holzapfelNonlinearSolidMechanics2010). This relationship, known as generalized Hooke's law, is expressed as:

$$\sigma_{ij} = C_{ijkl}\epsilon_{kl}. \quad (2.14)$$

For a, isotropic, linear elastic material, such as the snow in this case, the stiffness tensor simplifies and can be described by the Lamé-parameters. In the xx-direction this is:

$$\sigma_{xx} = (\lambda + 2\mu)\epsilon_{xx} + \lambda\epsilon_{yy} \quad (2.15)$$

(kAkiRichards19801981). When scaling this with the particle volume, the individual moment tensor components can be obtained using

$$M_{ij} = \Delta\sigma_{ij}V_p. \quad (2.16)$$

The scaling with the particle volume is done because in a small material volume, such as a snow particle, the stress drop is the same as a small seismic point source (backusMomentTensorsOther1976). This is why multiplying the stress tensor with the particle volume yields the moment tensor component.

These moment tensor components can then be input into SALVUS as custom source time functions. The waveforms are obtained via the amplitude of the different moment tensor components at different times.

The visual representation fo this workflow can be seen in appendix ??.

2.6 Simulation of DAS data

To compare the results obtained in the SALVUS simulations to field DAS data, the numerical outputs have to be transformed. This is because field DAS measurements, the output are the fields averaged over the gauge length (lindseyFiberOpticSeismology2021) rather than output at every point along the calble. The backscattered delay of a laser pulse along a straigh cable is proportional to the integrated strain tensor components along the gauge length (noeFrontiersFullwaveformSeismology2025). The relationship between the deformation tensor F and the phase delay of the laser pulse is described by:

$$\phi_{laser}(t) \propto \int_{-\frac{1}{2}}^{\frac{1}{2}} |I + F(s, t)e(s)|ds, \quad (2.17)$$

according to fichtnerTheoryPhaseTransmission2022. In this equation, I is the identity matrix and e is the local unit tangent vector at the arc length s along the DAS fibre in 3D (noeFrontiersFullwaveformSeismology2025).

Because the deformations of interest are small, meaning that $F \ll I$, equation ?? can be linearized. This leads to

$$\phi_{laser} \propto \int_{-\frac{1}{2}}^{\frac{1}{2}} e^T(s)\epsilon(s, t)e(s)ds, \quad (2.18)$$

where the strain tensor is $\epsilon = \frac{1}{2}(F + F^T)$ (fichtnerTheoryPhaseTransmission2022).

noeFrontiersFullwaveformSeismology2025 state that DAS interrogators scale phase delay measurements to convert them to strain, which leads to the vanishing of the proportionality. Equation ?? becomes

$$\bar{\epsilon}_{DAS}(s_c, t) = \frac{1}{l} \int_{s_c - \frac{1}{2}}^{s_c + \frac{1}{2}} e^T(x)\epsilon(s, t)e(s)ds \quad (2.19)$$

(noeFrontiersFullwaveformSeismology2025). This means that the measured strain is averaged over the gauge length l (noeFrontiersFullwaveformSeismology2025). Physically, this means that the strain field recorded by DAS is smoothed because of the averaging over the gauge length.

Methodology

3.1 Model Setups

3.1.1 Simple Model

A 2-layer domain spanning 400m in the x-direction and 3m in the y-direction was set up. This is the same domain as the coheisve zone model. The domain consists of a 1.5m snow layer and a 1.5m air layer, the parameters for this medium are listed in appendix ??, table ?. The medium extent of 400m was chosen to ensure that sources can be placed far away from the absorbing boundaries, but to still have a source line which would be physical as an avalanche release zone. Furthermore, a large domain combined with a long simulation time ensures that both the near- and the far-field of the wave can be observed, which is important for interpretation of field DAS data. In the simple case, a series of SALVUS moment tensor sources are set up with a certain distance, they span from 30m to 270m with a spacing of 10cm. The small source spacing was chosen to mimick a moving, continous source like a crack in snow, accurately (madariagaDynamicsExpandingCircular1976). The model domain setup can be seen in figure ?. The air layer is shown in light blue, the snow layer in dark blue. Because the sources have 10cm spacing, the source extent instead of individual sources is shown in red. The time delay with which the sources fire is dependend on the sub-rayleigh or supershear case which is implemented, the bounds for the crack propagation velocity are taken from trotterTransitionSubRayleighAnticrack2022, they are 0.6 or 1.6 times the shear wave velocity for sub-rayleigh and supershear cases respectively.

The bedrock or soil is not considered in both models. This is because preliminary tests (research_module) of a 3-layered medium have shown that it acts as a reflector of waves. This is because of the high impedance contrast between snow and bedrock, which causes waves to be reflected almost entirely at the interface kAkiRichards19801981. For computational efficiency, the bedrock layer was thus neglected and the bottom of the snow was made a regular boundary, in other words, reflecting, to mimick the presence of a layer below the snow. A reflecting bottom boundary is physically equivalent to a rigid half-space with infinite impedance contrast, which is a reasonable approximation given that snow P-wave speeds (109 m/s) are substantially lower than typical bedrock values (1500-3000,m/s) (kAkiRichards19801981). The top as well as both side boundaries of the model are set to be absorbing boundaries, which means that waves that reach these boundaries are absorbed and not reflected back into the model. This is done so that the model accurately represents the physical behavior of wave propagation in a much larger domain, which is the reality of field DAS data. The absorbing boundaries were set to be 3.5 times the maximum wavelength present in the model, this is according to the SALVUS documentation (afanasievSalvusFlexibleHighperformance2017).

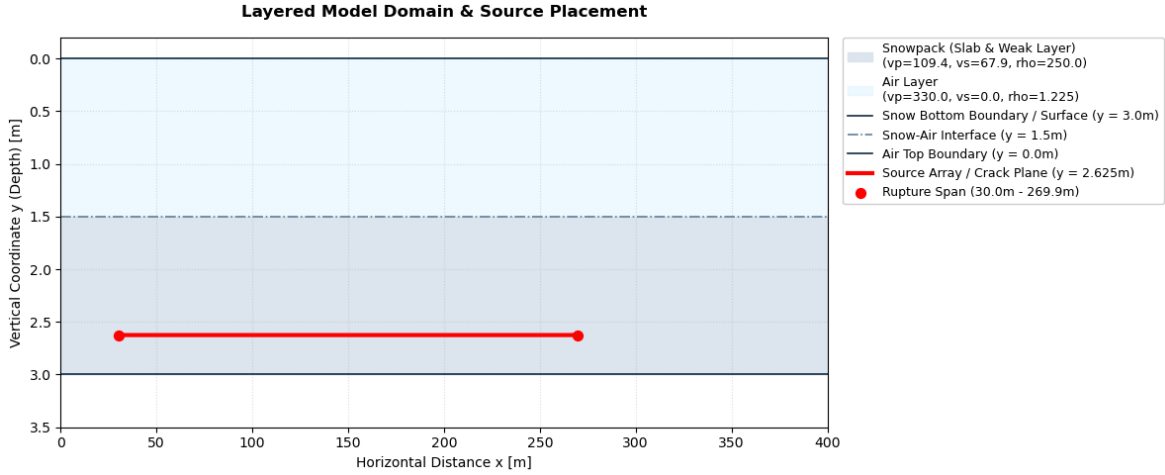


Figure 3.1: Layered model setup. Air layer in light blue, snow layer in darker blue. Source line extent in red. Model properities used can be seen in legend, but also appendix ??

3.1.2 MPM Model

To represent the results of the more complex source time functions obtained from the material point method simulations, the same layered model as trotterTransitionSubRayleighAnticrack2022 was implemented. This model is more complex than the two-layer case described in order to have results that represent avalanche physics better. The material properties, taken from trotterTransitionSubRayleighAnticrack2022, are listed in appendix ??, table ??. The p- and s-wave velocities are computed from the material properties via the shear and elastic moduli:

$$M_{shear} = \frac{E}{2(1 + \nu)}, \quad M_{elastic} = \frac{E}{(1 - 2\nu)(1 + \nu)} \quad (3.1)$$

(kAkiRichards19801981). From these, the wave velocities can be derived:

$$v_p = \sqrt{\frac{M_{elastic}}{\rho}}, \quad v_s = \sqrt{\frac{M_{shear}}{\rho}} \quad (3.2)$$

(kAkiRichards19801981).

This consists of the bed layer, a weak layer and a snow slab layer. In order to represent a physical snowpack with a weak layer buried in snow (schweizerSnowAvalancheFormation2003), the weak layer here is in the middle of the slab layer. To account for the absorbing boundaries which are implemented at the top and sides, the model domain was extended. At the sides, this was done by extending the model extent to accomodate the abcs. At the top however, an air layer was added which accounts for the abcs in thickness. This was done to keep the original thickness of the snow slab layer described by trotterTransitionSubRayleighAnticrack2022 but still have a boundary which leads to no reflections from the top of the domain. In other words, adding an air layer above the slab allows the snow-air free-surface boundary condition to remain at the correct physical height while the absorbing boundaries occupy the air layer above, which mimicks a domain with infinite extent in y while also avoiding any interaction between the ABC and the elastic snow medium (afanasievSalvusFlexibleHighperformance2017). The setup can be seen in figure ??.

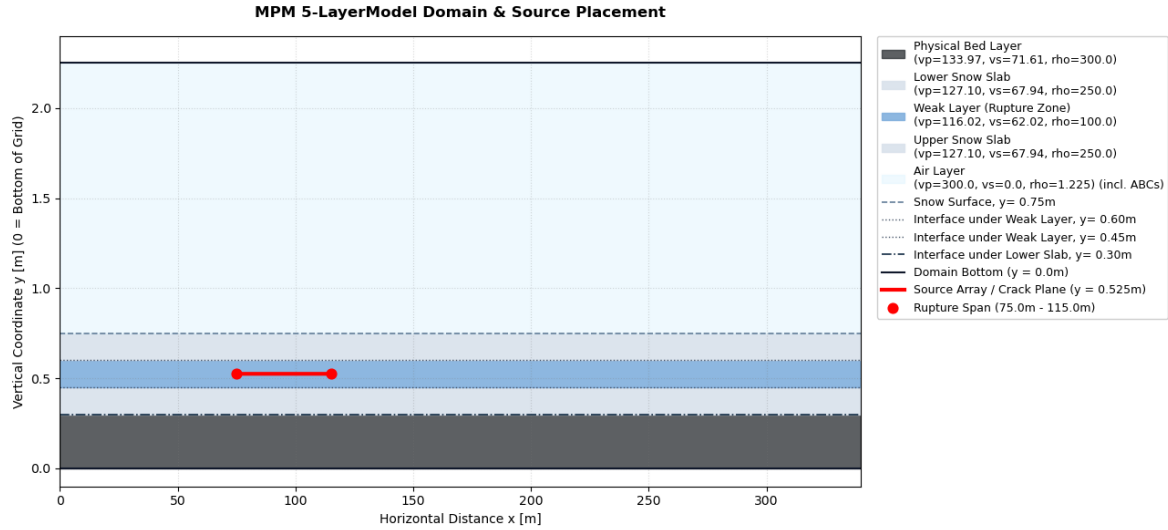


Figure 3.2: Layered model setup for MPM. Air layer in light blue, snow slab in light grey, weak layer dark grey and bedrock in blue grey. Source line extent in red. Model properities used can be seen in legend, but also appendix ??

3.2 Source Generation for Simple and CZM Model

All sources generated use SALVUS' custom source time function option. For the simple model and the cohesive zone model, this is done by writing a file with the source wavelets in them. The source wavelets have ricker shape with peak frequency f_m according to kAkiRichards19801981; graffWaveMotionElastic1975; RickerWavelet2021:

$$f(t) = (1 - 2\pi^2 f_m^2 t^2) e^{-\pi^2 f_m^2 t^2}. \quad (3.3)$$

The ricker wavelet was chosen over other source time functions in SALVUS because its frequency content is controlled at a single value, the central frequency (kAkiRichards19801981).

The workflow of the code, including how the source speed is set up can be seen in the following flowchart.

3.2.1 Cohesive Zone Model

A graphical representation of the model setup and the equations used can be found in appendix ??.

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following mahajanModelingInterfacialCrack2008. This model was chosen because there is a narrow band ahead of the fracture zone, the cohesive zone, where the material is already weakened. This means that prior to the failure of a point in the material, there is already weakening of the material. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer. Material parameters for the snow layer are taken from guillemotEffectSnowfallChanges2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

Each activated subfault is represented as a point source in the SALVUS source array. The kinematic moment formulation serves as the numerical link to the underlying cohesive zone model through a

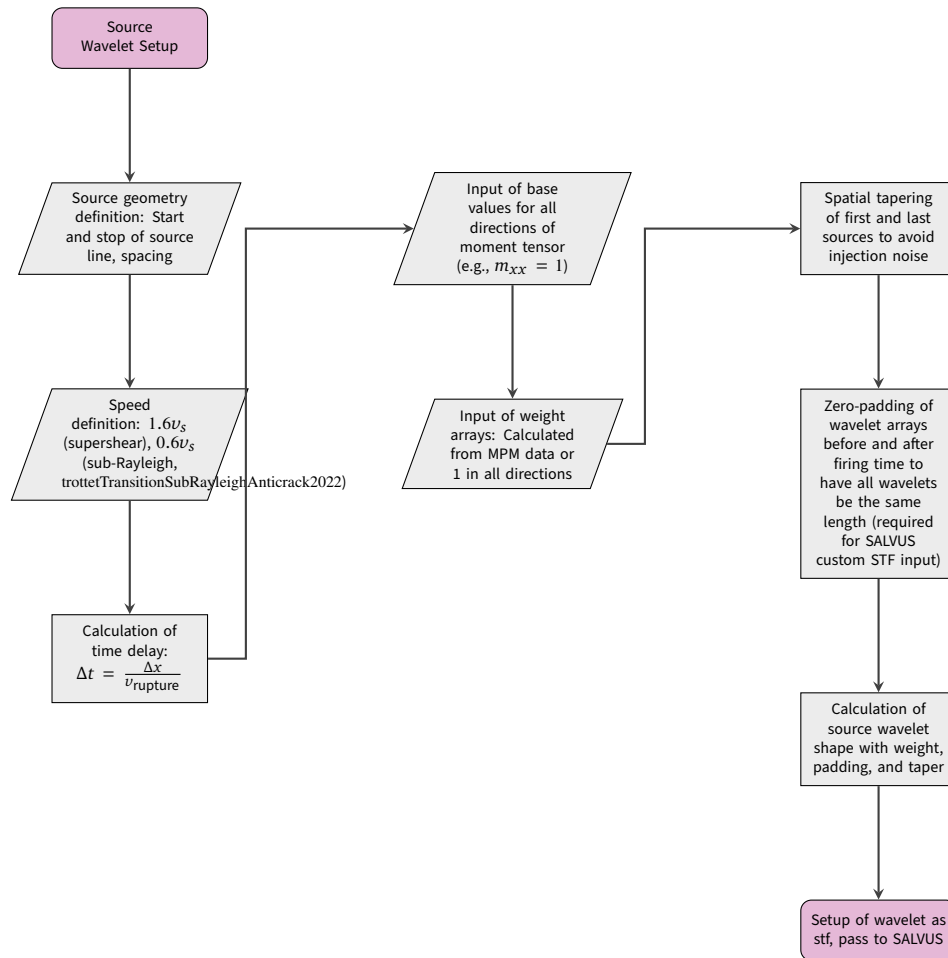


Figure 3.3: Operational workflow for the SALVUS source time function (STF) setup.

strict energy balance. Within the CZM, the total strain energy released during fracture is split between the fracture energy (G_I or G_{II}) required to break down the cohesive zone, and the remaining radiated elastic strain energy. By prescribing x_{slip} across a finite grid cell width Δx , the resulting scalar seismic moment M_0 mathematically captures the radiated energy that propagates away from the process zone into the bulk medium as a seismic wavefield. The scalar seismic moment per subfault is

$$M_0 = \mu \delta_c \Delta x, \quad (3.4)$$

where μ is the shear modulus of the weak layer, δ_c is the critical slip distance, and Δx is the subfault spacing.

To take the mixed-mode mechanics of the process zone into account, the individual components of the moment tensor are scaled by the mode-mix ratio r_{mode} , which can be user input:

$$m_{xx} = 0, \quad m_{xy} = M_0 r_{\text{mode}}, \quad m_{yy} = -(1 - r_{\text{mode}})M_0.$$

Consequently, at the extremes of $r_{\text{mode}} = 0$ or 1, the unaligned component of the moment tensor drops to zero, perfectly matching the radiation patterns of pure opening or pure shear failure. All subfault sources are driven by an identical central frequency, source wavelet, and amplitude scaling factor, thereby reconstructing the discrete subfault array into a highly coherent, continuously moving source that mimics a smooth, physical crack tip rupture throughout the wave propagation mesh.

Key parameters and their provenance used in the simulations are summarized in appendix ??, table ??.

3.3 Source Generation for MPM

For the MPM results that are translated to FE, the moment tensor components and source wavelet shapes are obtained as described in section ?. These are then loaded and translated into individual components, which can be passed to SALVUS. The time delay previously described is already contained in the MPM data. A visual representation for the workflow with the MPM source wavelets is:

The directionality and amplitude of the moment tensor components extracted over all sources can be seen in figure ?. The colorbar shows the amplitude of the individual components. This is important because the source components show the predominant directions of the sources, which is important for the interpretation of the wavefield. The diagonal boundary separating zero (blue, lower triangle) from active values (upper region) marks the crack propagation front. The moment tensor components are injected into SALVUS as custom source time functions, plotted as a function of source index (horizontal axis, corresponding to spatial position along the crack) and MPM timestep (vertical axis).

From all figures ?, it can be seen that the moment tensor has a predominantly negative direction that does not bounce back to neutral. To further investigate this, the cumulative integral of the moment tensor integrated in respect to space was calculated:

$$I_{ij}^{\text{tot}}(t) = \int_0^t M_{ij}^{\text{tot}}(\tau) d\tau, \quad (3.5)$$

which if nonzero, means that there is a residual, non-reversible deformation. This results in figure ?, which supports the trend seen in figure ?, that there is a negative permanent deformation in respect to y after the sources have stopped firing.

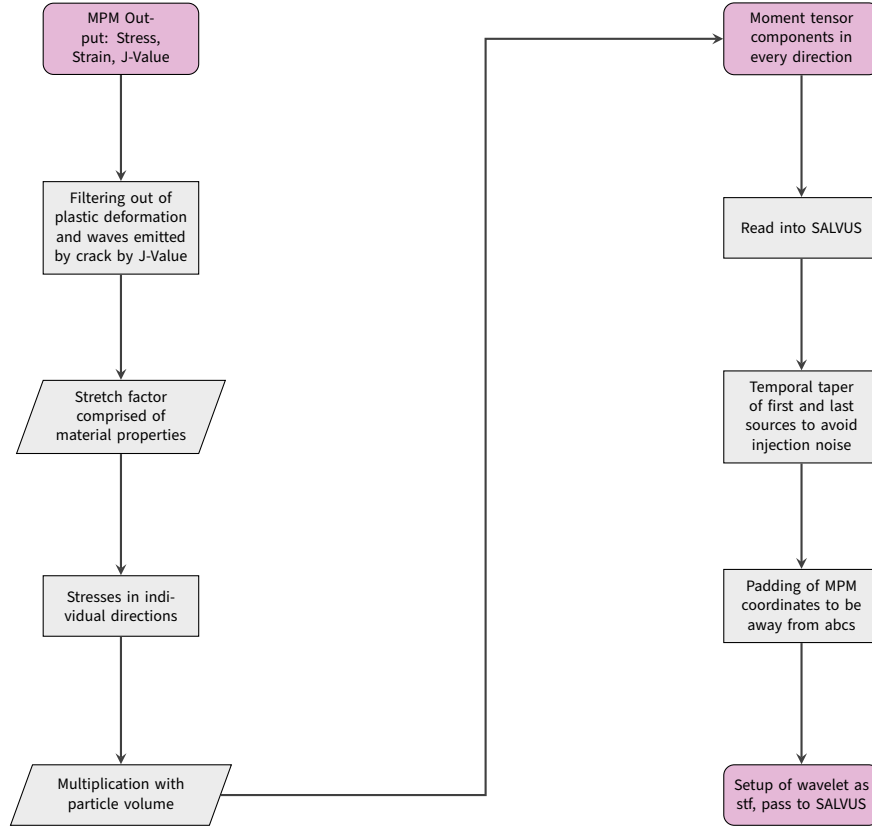


Figure 3.4: Workflow for conversion of MPM data and input as SALVUS STF.

3.4 Simulation of DAS data

The theory in section ?? is used to simulate DAS data from the outputs given by SALVUS. This is done by extracting the displacement output. The spatial derivative of displacement is taken to obtain strain: $\epsilon = \frac{\partial u}{\partial x}$, the temporal derivative of strain results in strain rate ($\frac{\partial^2 u}{\partial x \partial t}$).

Strain and strain rate are averaged over a 10.4 m gauge length, which was chosen based on the avalanche warning system installed by turquetAutomatedSnowAvalanche2024, which has a gauge length of 10.4m. This is combined with the Pyber library by noeFrontiersFullwaveformSeismology2025. The pyber gauge length physically averages the data over the input gauge length, which is the same as what is described in section ?. This is done by calculating weights for the points within the gauge length window, all weighted points in a window are then added up (noeFrontiersFullwaveformSeismology2025). To replicate smoothing caused by the DAS laser pulse, pyber uses a Gaussian window, which calculates the weights. A Gaussian window is used instead of a rectangular (boxcar) window to more accurately represent the Gaussian shape of the interrogator laser pulse, which produces a weighted rather than uniform average over the gauge length (noeFrontiersFullwaveformSeismology2025).

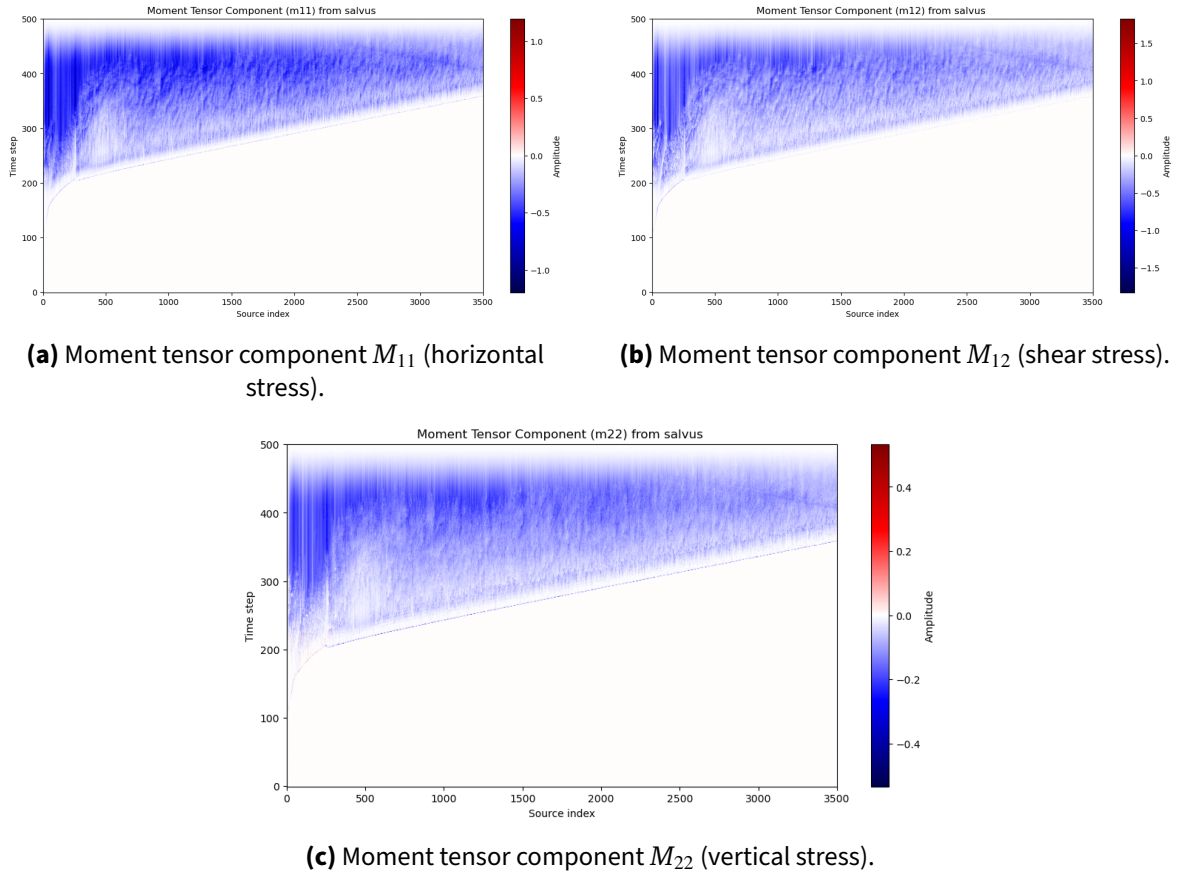


Figure 3.5: Moment tensor components injected into SALVUS as custom source time functions, plotted as a function of source index (horizontal axis, corresponding to spatial position along the crack) and MPM timestep (vertical axis). The colorbar shows amplitude (blue: negative amplitude, red: positive amplitude). The diagonal boundary separating zero (blue, lower triangle) from active values (upper region) marks the crack propagation front.

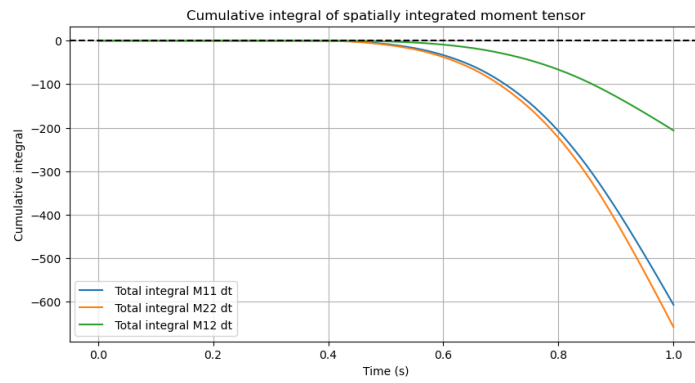


Figure 3.6: Cumulative integral of moment tensor components. Blue is m11, green is m12 and orange is m22. X-axis shows time, y-axis shows magnitude of cumulative integral.

Results

4.1 Simple Model Results

The following results shown are for the supershear and sub-rayleigh case of the custom stfs. The sources are spaced 0.1m apart between 30m and 270m and have a ricker wavelet all directions. The moment tensors have components 1 in xx and xy and -1 in yy. The simple, two layer model was set up to test the custom STF option in SALVUS and to verify how it works. The results are shown here because it is important to understand the physics underlying crack propagation in snow before moving on to the more complex case. Since the focus of this thesis is to model crack propagation as recorded by DAS data, strain and strain rate plots are shown first. The velocity results are shown to verify observations made in the strain and strain rate plots. The results of the CZM model will not be discussed in detail. This is because the simple model results suffice to understand the MPM-aided model results. Additionally, if physics of the CZM model are kept simple, for example when there is no mode mix of the fracturing, the CZM results are very close to the simple model results. Nevertheless, setting up this model was important to understand snow mechanics and crack propagation physics.

4.1.1 Sub-rayleigh strain and strain rate plot

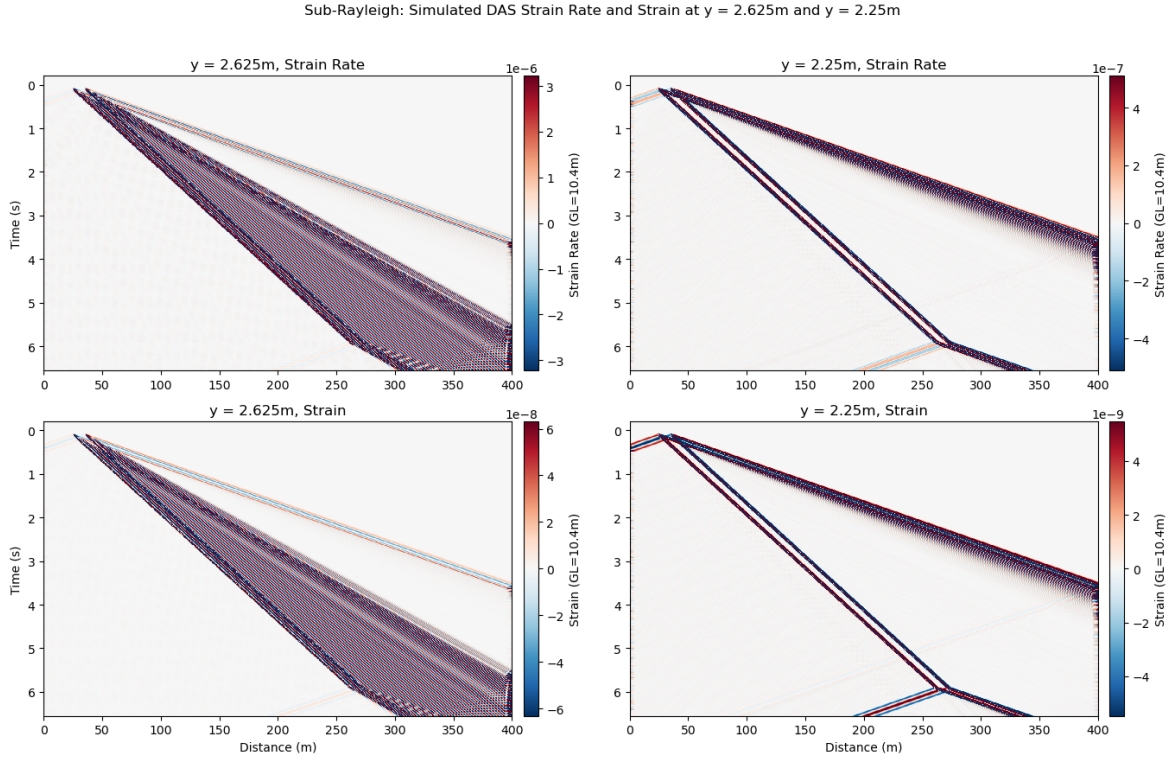


Figure 4.1: Sub-rayleigh strain and strain rate (horizontal component) averaged over 10.4 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

Both the strain and the strain rate at crack depth look very similar to each other, the same is true for the snow bottom. At crack depth, both the strain and strain rate in the left column of figure ?? show one strong main front, which is preceded by a preliminary front. The main front with higher amplitude has a slope which corresponds to the prescribed rupture speed, whereas the precursor is faster and has lower amplitude (fainter color). The same is true for the snow bottom, but here, the main front is thinner than the preliminary front. The main front still has higher amplitude than the preliminary front, but the preliminary front weakens in amplitude with increasing time.

At crack depth, the main front broadens with increasing time, this can not be observed for the main front measured at the snow bottom on the right of figure ?. At both depths, for strain and strain rate, the two fronts have visibly different slopes, which corresponds to different apparent propagation speeds.

At both depths, a continuous wavefront extends from 0s starting at 30m to 6s ending at 270m, this corresponds to the source coordinates defined and also to the time delays prescribed to the sources.

In all plots, the strains and strain rates following each other have opposite polarity. The magnitude of strain and strain rate is larger at the crack depth compared to the bottom of the snow. This is indicated by the colourbar range at the snow bottom being approximately one order of magnitude smaller than at crack depth. This indicates strong amplitude decay over the 0.375,m vertical separation.

At $x=270\text{m}$, $t=6\text{s}$ in all sub-rayleigh plots in figure ??, the main wavefront stops, but there is a new wavefront propagating from this point. This front has a change in slope, which means a different

propagation speed, which coincides with the rupture arrest front.

4.1.2 Supershear strain and strain rate plot

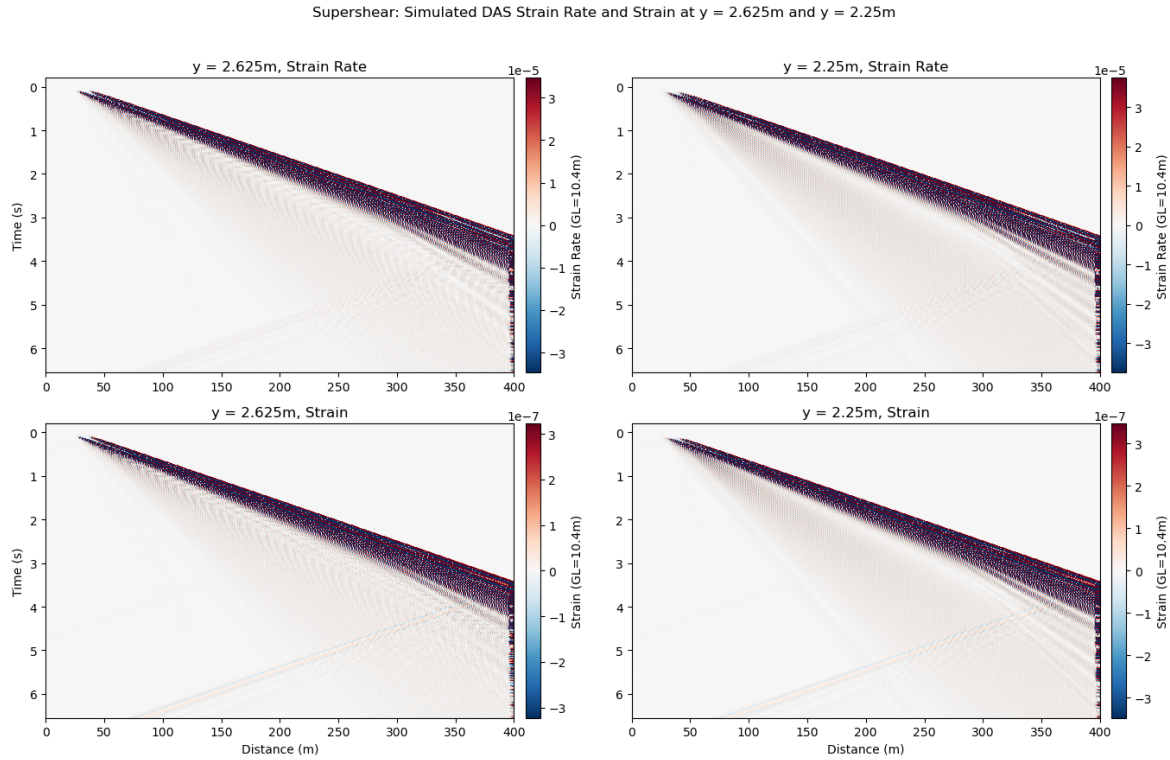


Figure 4.2: Supershear strain and strain rate (horizontal component) averaged over 10.4 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

In the supershear case, strain and strain rate at both receiver depths in figure ?? show a single dominant wavefront with comparable amplitude, with no significant reduction between crack depth and snow bottom. This main wavefront has strong amplitude at first, but experiences significant reduction in amplitude after a while. This corresponds to loss of energy of the different frequencies contained in the wavefront.

The strain and strain rates observed have opposite polarity (positive followed by negative). In the strain rate plots, the different polarities can be seen a bit more clearly than in the strain plots in the bottom row of figure ??.

In figure ??, there is no difference in the magnitude of strain and strain rate between the crack depth and snow bottom. However, the peak strain rate in the supershear case is approximately one order of magnitude larger than in the sub-Rayleigh case, as indicated by the colourbar scales in figures ?? and??.

Similarly to the sub-rayleigh plots in figure ??, the supershear plots also show wavefronts with different amplitudes at the rupture arrest at 270m. Here, this is characterized as a fan of distinct wave trains coming from the rupture arrest point.

4.1.3 Sub-Rayleigh Velocity plots

The following plots show horizontal (v_x) and vertical (v_y) particle velocity at two receiver depths: 2.25m, which is the snow bottom, and 2.625m which is the depth of the crack. The particle velocity shown is derived from the displacement field, this is done to assimilate the results shown more to DAS field measurements. In DAS field data, the instantaneous particle velocity at one point is not measured, but rather averaged over a gauge length (lindseyFiberOpticSeismology2021). By taking the temporal derivative of displacement, the results shown here are closer to field results.

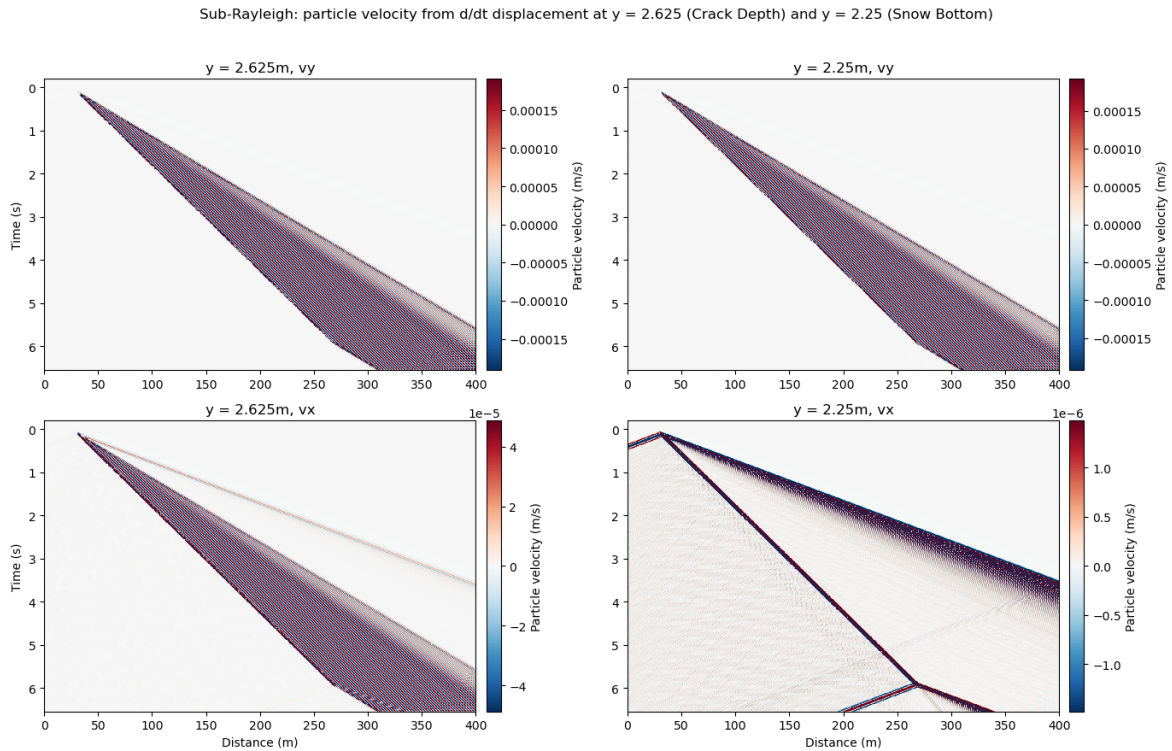


Figure 4.3: Sub-Rayleigh Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

The sub-rayleigh velocity plots in figure ?? all show a distinct wavefront starting at 0s at 30m and ending at 270m. This main wave train ends at 6s, but sends out waves with a different speed (different angle) in direction of the source propagation. This corresponds to the arrest of the rupture which was already observed in the strain and strain rate plots. The horizontal velocity at the snow bottom also shows a backward-propagating train from the crack tip.

Both v_x plots (bottom row in figure ??), show the main wavefront with waves travelling ahead of it, as time and distance progresses, the main wavefront gets wider. This means that they have different slopes, which corresponds to different propagation velocities. In all but the plot of v_x at the snow bottom, the first couple of waves in the main wavefront are weaker in color, and thus weaker in velocity. In v_x at the crack depth at the bottom left of figure ??, the precursor is weaker in color and thus magnitude of velocity than the main wavefront. At the snow bottom, which is shown on the bottom right, the precursor has a high amplitude of velocity initially, which then decays.

In the main wavefront observed, it's very noticeable that the particle velocity is layered. In other words, there are stripes along the wavefront with opposite polarity, for example, there is a stripe with positive polarity (red) followed by one of negative polarity (blue). This is also noticeable in the cases that have multiple wavefronts, although it's less pronounced.

In all panels of figure ??, the main wavefront between 30m and 270m does not decay in amplitude with increasing time. It is noticeable however that in all but the horizontal velocity at the snow bottom, the main wavefront broadens with time, which corresponds to dispersive wave propagation in a bounded medium.

4.1.4 Supershear Velocity Plots

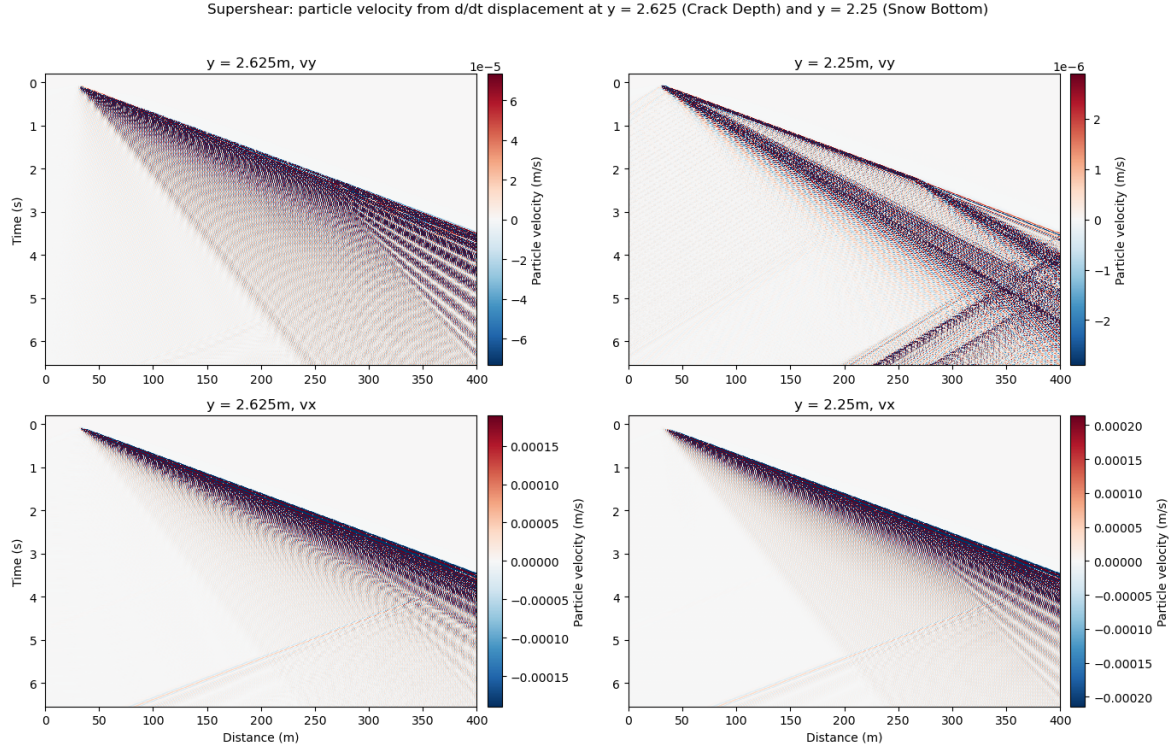


Figure 4.4: Supershear Particle velocity: vertical velocity in top row, horizontal in bottom row. Crack depth in left, snow bottom in right column. Colorbar is unscaled, showing velocity (m/s).

In all of the supershear velocity plots in figure ??, a single dominant wavefront extends from 30m to the end of the medium at 400m. The slope of this front corresponds to the prescribed supershear rupture velocity. This wavefront broadens with increasing propagation time, where it also loses amplitude (weaker color). At the crack arrest at 270m, the end of the source array, the wavefront generates a fan of outward-radiating waves consistent with stopping-phase radiation from the crack arrest point (madariagaHighfrequencyRadiationCrack1977).

In the v_y panel at the snow bottom (top right in figure ??), there is a clear distinction between two wavefronts, which cannot be seen in all other supershear velocity space-time plots. The first wavefront with high amplitude corresponds in slope and therefore propagation speed to the wavefront in all other plots, it also has a fanning out of waves from 270m, so the rupture arrest. From the crack onset at 30m, there is a second wavefront which also is of a higher amplitude. This wavefront has a shallower slope and therefore a lower propagation speed than the main wavefront and does not emit a fan at the rupture arrest point.

The wavefronts arriving in all cases switch between polarities. While the pattern of polarity is the same in all cases, the slope of the wavefronts change. In all plots, after the first wavefront, the waves seem to slow, which means their slope flattens.

4.1.5 Frequency-Wavenumber (fk) Plot

The fk-plot in figure ?? shows the energy (colorbar) density at frequencies (y-axis) and wavenumbers (x-axis) present in the results. This energy was taken from the horizontal velocity results at the snow bottom because it corresponds to the data measured by a horizontal DAS cable buried in snow. The plots shown are normalized to the 99th percentile of the energy present, this is done so that outliers with large energies are scaled and do not skew the color scale.

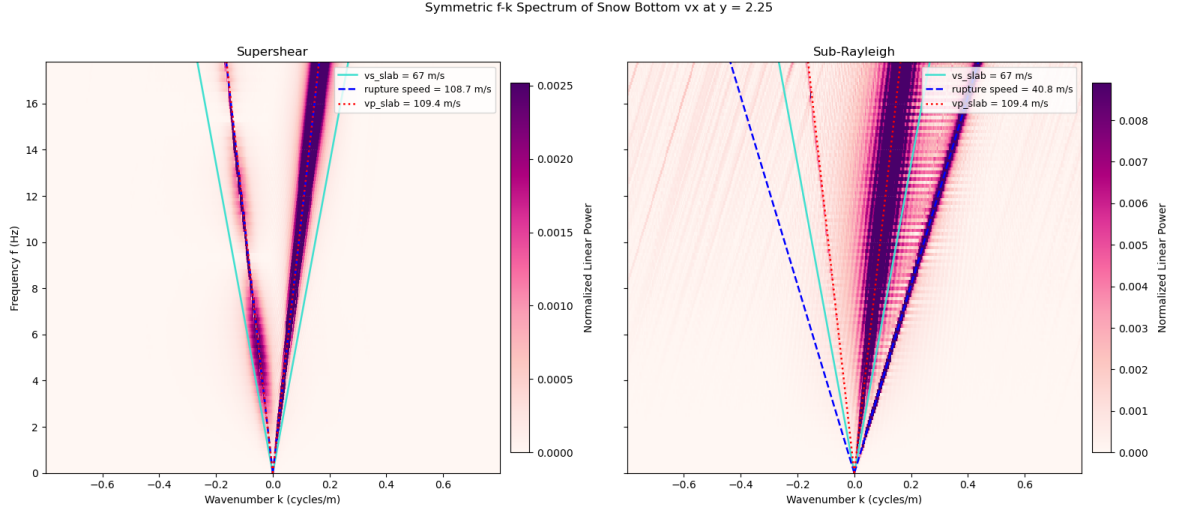


Figure 4.5: F-k plot for the supershear (left) and sub-rayleigh (right) case. Slab p-wave velocity in red, s-wave velocity in turquoise and respective rupture speed in blue. Colorbar shows energy normalized to the 99th percentile.

The supershear and sub-Rayleigh f-k plots in figure ?? show qualitatively different energy distributions, consistent with the two distinct propagation regimes. The supershear case shows a concentration of energy around the rupture speed (blue line) and the vp (red line) of the slab on both sides of the wavenumber spectrum. It is important to note here that these speeds are very close in value, so the lines cannot be distinguished well in the plots. The sub-rayleigh case also shows two concentrations of energy. A very broad band at the vp of the slab and a concentration of energy at the rupture speed on the positive wavenumber side. In all cases, the energy is more concentrated around the vp and or rupture speed for lower frequencies, at higher frequencies, the concentration of energy starts to deviate from speeds present in the medium in figure ??.

In the supershear case, the dark spots of concentrated energy are more broad on the positive side compared to the negative side, where they are narrower and less smeared out. This asymmetry reflects the directionality of the source: the crack propagates in positive x and more energy is released in that direction. In the sub-rayleigh case on the right side of figure ??, there is no concentration of energy at negative wavenumbers.

4.2 MPM Guided Results

The results shown here are from the MPM-guided simulation. The sources are spaced 0.01m apart from 75m to 125m. Unlike the results in section ??, which are for a simple, 2 layer model with prescribed rupture velocities (sub-rayleigh and supershear case), the results here are for a 5-layer model with a rupture velocity coming from the MPM simulations. The results shown here are filtered using a low-pass filter with 50Hz cutoff. Because both strain and velocity are derived from displacement,

the displacement field extracted is shown in appendix ???. The displacement data itself is unfiltered, but because strain, strain rate and velocity are derivatives, the differentiation acts as a high-pass filter which amplifies noise. Displacement is shown in the appendix to show that there are little numerical artefacts and noise in the extracted data. This means that filtering strain, strain rate and velocity should not be a concern because the noise is simply from the derivative.

4.2.1 MPM Strain and Strain rate

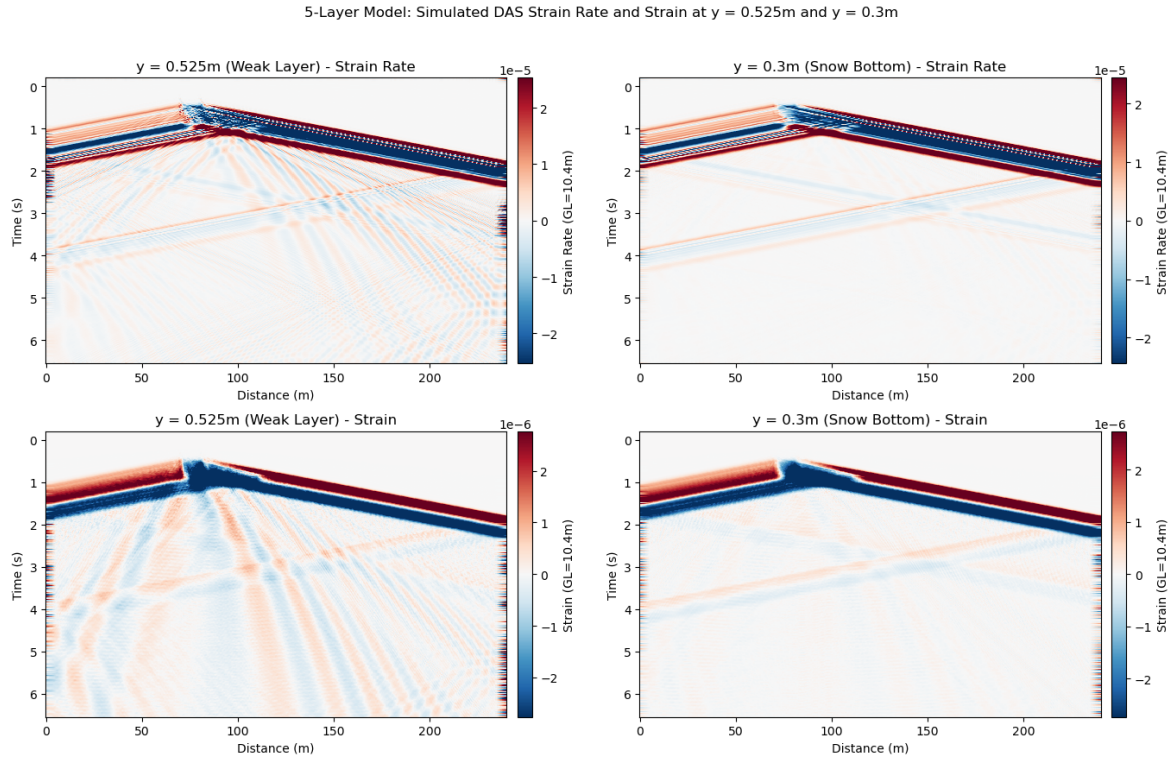


Figure 4.6: Strain and strain rate (horizontal component) averaged over 10.4m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left, snow bottom on the right. Colorbar shows magnitude of strain or strain rate.

The strain as well as the strain rate plots in figure ?? all show a distinct v-shape initiating from the crack onset point at $x=75\text{m}$. The left arm of the V corresponds to backward-propagating body waves radiated from the crack initiation point, while the right arm corresponds to the propagating crack traveling from the initiation point onwards.

Both strain panels (bottom column) show a front of positive strain (red, on top) arriving first, which is followed by a negative strain front. The extension (positive strain) followed by compression could indicate P-waves or the S0 Lamb wave mode. At both the crack depth as well as the snow bottom, in positive x-direction, the red front widens until about $x=125\text{m}$, when there is a jump and the front has about same width as the blue front. This location coincides with the crack arrest location. In the negative x-direction, both the positive and negative front have the same width from the initiation point on to the domain boundary.

The same positive and negative fronts from the crack onset point can be observed in the strain rate plots in the top row of figure ?. Contrary to the strain panels, where the fronts are solid colors, in the strain rate panels, these are much more heterogenous. The forward propagating positive front is much narrower in strain rate than in strain, the same is true for the backward propagating negative front. The wedge-shape to the crack arrest point observed in strain can still be seen in strain rate

however. In the strain rate plots there also more than just two wavefronts in the v-shape coming from the crack onset. The main positive and negative fronts are still visible, but there is another positive front under the blue front in the positive direction. In the negative direction, the same is true: under the blue wavefront there is multiple positive and negative layered wavefronts.

In the strain plot at crack depth as well as in the strain rate at the same depth, there are more wavefronts additional to the v-shape from the crack onset. Those fronts fan out from the crack initiation point at 75m as well, but they have a much steeper slope than the main front. In contrast to the main front, they are weaker in color meaning amplitude. These fronts are layered in polarity, meaning a front of positive polarity (red) is followed by one of negative polarity.

These weaker amplitude fronts broaden with increasing time, which is consistent with dispersive wave propagation. Their amplitude does not decrease however. In the strain rate at crack depth ($y=0.525m$, weak layer, shown on top right of figure ??), the colors of the second wavefront are weaker than in the strain panel at the same depth. This means that they have lower amplitude relative to the main wavefront in the strain rate panel.

The main wavefront observed in figure ?? in all panels does not broaden over time, it also shows no fading in color, which would indicate a decrease in amplitude.

In all panels, there are two red-blue wavefronts coming from the domain side boundaries. These are especially noticeable in the snow bottom strain and strain rate plot (right side of figure ??), because the main front is the only wavefront present there. The wavefronts coming from the side boundaries of the domain have the same slope as the main wavefronts (wavefront coming from the right boundary has the same slope as backward-propagating front from crack), but they are much weaker in amplitude.

When comparing the strain rate plots (top column) to the strain plots, although the main structures are the same, the colorbar scale in the strain rate plots is one order of magnitude (one power of 10) larger than that of the strain plots. This does not depend on depth (snow bottom has the same amplitude as crack depth) but on the type of field shown.

4.2.2 MPM Velocity

The velocity plots in figure ?? show that both cases look pretty similar to each other independent of depth. This means that the vertical velocity in the top row, looks almost identical at crack depth compared to the snow bottom. The vertical velocity plots can be distinguished from the horizontal velocity plots however.

The horizontal velocity in the bottom two panels of figure ?? both show a flat v-shape from the crack origin at 75m, similar to that observed in the strain and strain rate plots in figure ?. The forward propagating front is characterized by a wavetrain of negative velocity (blue), which is the first arrival, followed by a train of positive energy (red) which arrives shortly after. In the direction backwards of propagation, this order is flipped: a positive front arrives followed by a negative front. In the horizontal velocity plots in figure ??, the vertex of the fronts with opposite polarity seems to be shifted. The first front, which is blue in positive direction of propagation and red in negative direction, is at $x=75m$, which is the crack onset. The vertex of the second v-shape, which has opposite polarity (red in positive, blue in negative direction), is not right below the vertex of the first v-shape. It is shifted in positive direction of propagation, which indicates that it could come from the crack arrest point at $x=125m$.

From the same vertex, in the v_x plots at crack depth (bottom left in ??), fronts with weaker amplitude originate. These wavefronts fan out from around $x=125m$ and they are layered in polarity (positive followed by negative). They also widen as time increases, which could indicate dispersion. Such

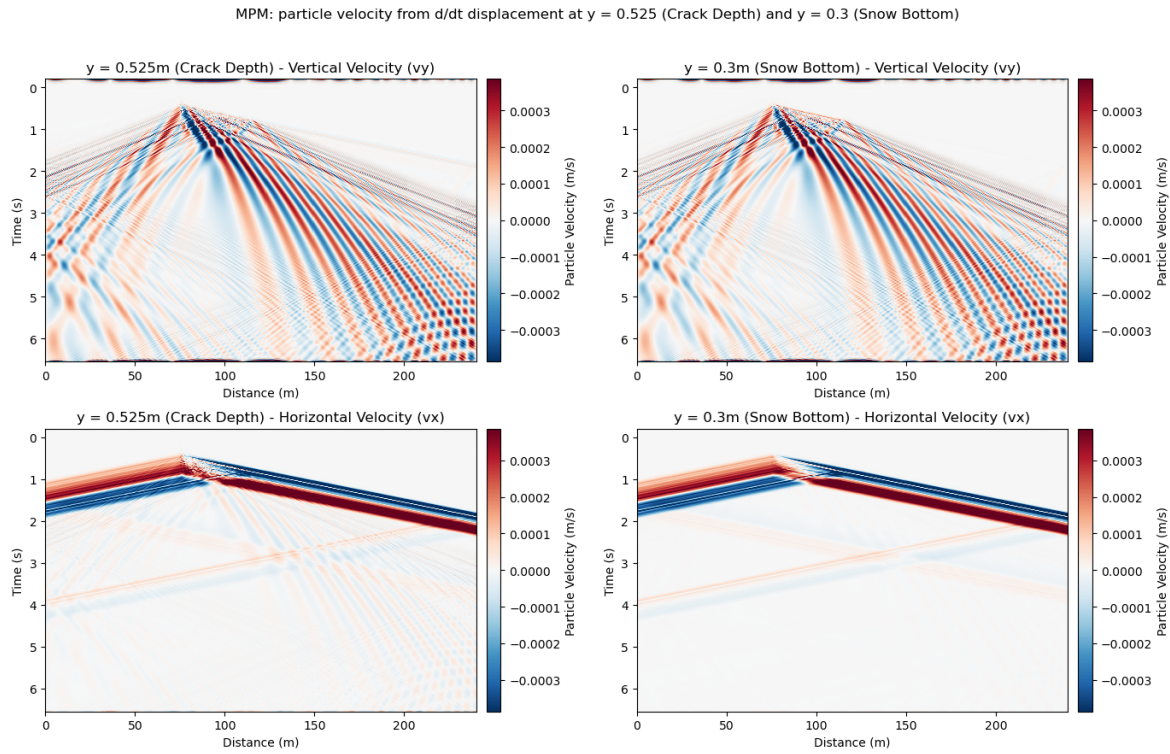


Figure 4.7: Particle velocity: vertical velocity in top row, horizontal velocity (v_x) in bottom row. Crack depth on left, snow bottom on the right. Colorbar shows magnitude of particle velocity (m/s).

secondary wavefronts are not present in the v_x plot at the snow bottom.

In the vertical velocity plots in the top row of figure ??, the secondary wavefronts are very strong and can be seen clearly at both depths. They originate in a v-shape from the crack initiation point at $x=75\text{m}$. These wavefronts are very strong in amplitude (saturated colors), and are layered in polarity. The wavefronts in both the positive and negative direction of propagation widen as time increases and their amplitude decreases, which is indicative of geometric spreading and dispersion. When looking at the later times in the top panels of figure ??, the blue and red velocities seem to curve towards the origin of the plot (towards $x=0$). This is another indicator for dispersion of higher frequencies.

Although the wavefronts from $x=75\text{m}$ are most noticeable, there are v-shaped wavefronts which also originate at $x=125\text{m}$ which is the crack arrest point. These are mainly noticeable at earlier times, for example between 1s and 2s, because at later times, there are multiple other wavefronts interfering with them. The second v of waves from the crack arrest starts later in time than the one from the crack initiation point, which is intuitive because the crack onset and arrest are not at the same time. The slope of the individual components from the arrest wavefront is similar to the one of the onset.

From the crack initiation point at $x=75\text{m}$, in the vertical velocity panels at the top of figure ??, there is a second wavefront which also propagates in the positive and the negative direction. The wavefronts in this front are much thinner than in the front that was described previously, they also have a flatter slope, which means lower speed. The slope of this wavefront is comparable to the one in the v_x plots below them. This wavefront also has layered polarity, but here the bands of one polarity are much thinner than in the v_x plots or the other wavefronts in the v_y plots.

In the area where the crack propagates, so between 75m and 125m, at the early times in v_y in the top panels of figure ??, the slope of the thinner layer wavefront is different than in the rest of the

wavefront. This area has a slope which is steeper than the rest of the wavefront, it is especially noticeable in v_y at the snow bottom, where a red line of positive velocity seems to connect the crack onset and arrest point. The slope of this line is likely the crack propagation speed.

In the vertical velocity plot at crack depth, one very faint wave from the crack onset point can also be seen. This wavetrain has a different slope than all other wavefronts in the plot, but it is very weak in amplitude (color).

4.2.3 MPM f-k plot

The data used for the f-k plot is the horizontal velocity data, for the reasons described in ??.

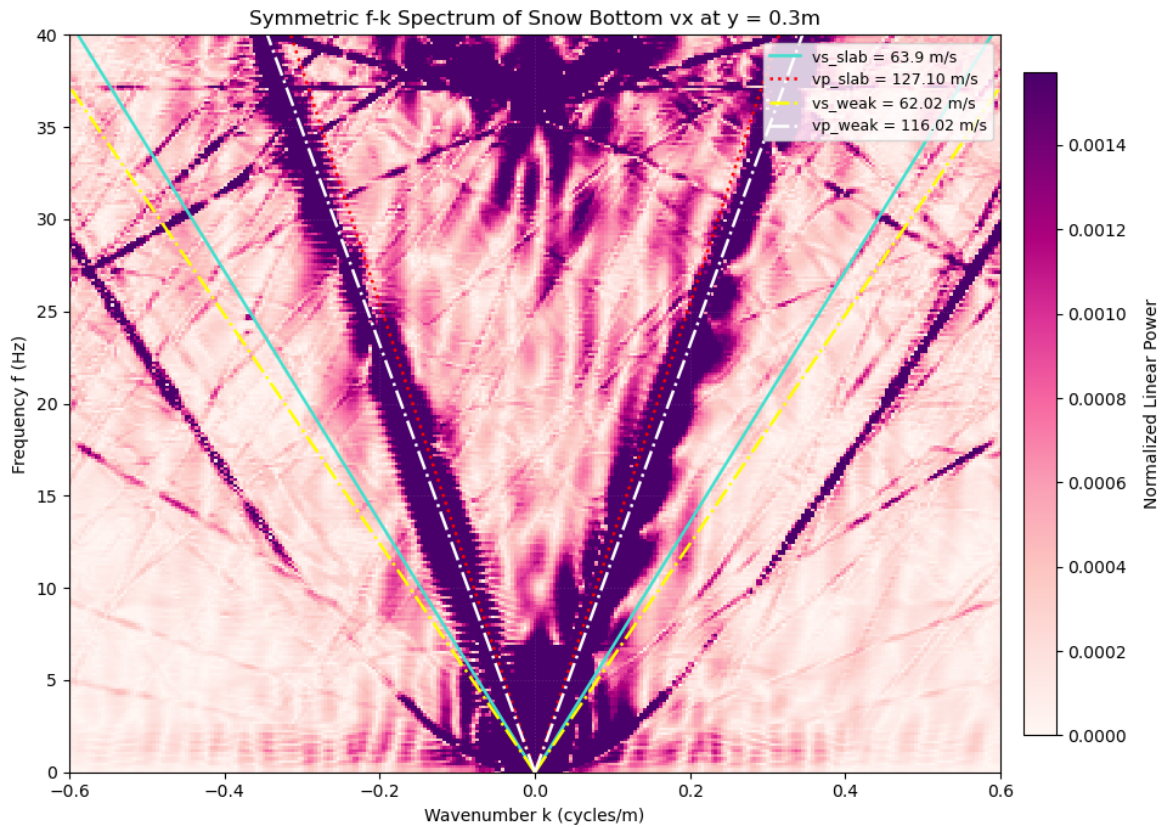


Figure 4.8: F-k plot for the MPM-guided simulation. Slab p-wave velocity in red, slab s-wave velocity in turquoise. Weak layer p-wave velocity shown by the white layer, s-wave velocity of WL shown by yellow line. Colorbar shows energy normalized to the 99th percentile to avoid outliers.

The frequency-wavenumber plot in figure ?? shows a symmetric distribution of energy around the origin. A lot of energy is concentrated around the positive and negative slab and weak layer p-wave velocity, which are indicated by the red (slab vp) and white (wl vp) lines.

The concentration of energy is not strictly at those two speeds, the dark spot of concentrated energy is broader and more spread out and therefore not contained by the two lines.

In the f-k plot in figure ??, there is no energy concentrated at the s-wave speeds of the slab, which is indicated by the turquoise line, or the v_s of the weak layer, shown by the yellow line.

Additional to the energy concentrated around the p-wave speeds, there are bands of concentrated energy on the positive and negative side of the wavenumbers. These energy bands are stronger in the positive direction and they have a shallower slope than the s-wave speeds present in the medium.

Discussion

5.1 Simple Model

5.1.1 Strain and strain rate space-time plots

The main wavefronts which can be observed in figures ?? and ?? are the crack fronts. This is because when taking the slope of these two wavefronts, it corresponds to the rupture velocity in the supershear and the sub-rayleigh case. The prescribed versus the calculated speed from the figures can be seen in table ?. Both cases produce strain and strain rate outputs that look very different, this is important when relating the results here to field measurements. The different behaviors mean that in theory, from the velocity output, the transition point between sub-rayleigh and supershear propagation regimes, as described by trotterTransitionSubRayleighAnticrack2022; bergfeldSignaturesSubRayleighSupershear2025; sironTheoreticalFrameworkDynamic2023, could be observed because the strain and strain rate measurement would change.

Case	Prescribed Speed	Speed from Results
Sub-Rayleigh	$0.6v_s = 40.8\text{m/s}$	$\frac{270-30\text{m}}{6\text{s}} = 40\text{m/s}$
Supershear	$1.6v_s = 108.7\text{m/s}$	$\frac{270-30\text{m}}{2.3\text{s}} = 104\text{m/s}$

Table 5.1: Simulation input rupture speeds versus speeds calculated from slopes in figures ??, ??.

Two-wavefront structure in Sub-Rayleigh

In the sub-rayleigh strain and strain rate plots, two main wavefronts can be observed. One, the main wavefront, which is broadest in the strain rate and strain plots at crack depth in figure ??, corresponds in slope to the sub-rayleigh rupture speed, as discussed above. The second wavefront is faster and arrives sooner than the main wavefront and only appears to be emitted from the first source. To confirm that this precursor is a physical wave type rather than a simulation artefact, simulations with larger source spacing were run and showed this changes when source spacing is increased, for example from 0.1m to 10m, as shown in appendix ??, figure ?. This leads to the conclusion that the precursor to the main wavefront disappears because of interference. All other sources but the first and the last one have interference from the sources before and after it, which means that this wave type, although emitted, is cancelled out because of interference (madariagaDynamicsExpandingCircular1976). When the source spacing is increased, the fronts of the individual sources travel further apart in space and they interfere less.

This precursor in the sub-rayleigh strain and strain rate plots has a slope which corresponds to a speed around the p-wave speed in the snow, which is also close to the speed of the symmetric lamb wave mode (S0). The reason for this is discussed in the velocity section ?? because it is easier to derive from the velocity output.

Wavefront identification and polarity sequence

In both the sub-Rayleigh (figure ??) and supershear (figure ??) cases, all panels show a consistent polarity sequence: positive strain (extension, red) arrives first, followed immediately by negative strain (compression, blue), then a return to positive. This extension-first, compression-second sequence is the expected signature of a compressional P-wave or S0 Lamb mode arrival at a receiver ahead of the source, and is well established in elastic wave propagation theory (kAkiRichards19801981). When the crack-tip stress field expands into the undisturbed medium, it first pushes material away from the source (extension), and the elastic restoring force of the medium then pulls it back (compression) before the wavefield equilibrates. The same sequence has been reported in field measurements of avalanches with geophones where p-wave precursors have been observed (herwijnenSeismicSensorArray2011).

After this leading extensional arrival comes the dense alternating polarity pattern of the A0 flexural wave train. As derived in section ??, the horizontal strain of the A0 mode varies as a cosine in space and time, cycling between maximum tension and maximum compression over each half-wavelength. The detailed arguments of why the A0 Lamb wave mode is present in the results is discussed in the velocity section ??.

Rupture Arrest Structure

As observed in all plots in section ??, the stopping of the crack at $x=270\text{m}$ produces a different wave radiation pattern compared to the crack itself. In the supershear case in figures ??, this manifests as a fan of waves coming from the crack endpoint, whereas in the sub-rayleigh case, there seems to be a v-shape coming from the tip of the crack.

The fan in the supershear case is consistent with stopping-phase radiation from a crack that has outrun its own shear wave cone (madariagaHighfrequencyRadiationCrack1977). Because the rupture speed exceeds the S-wave speed, the accumulated wave energy can only release forward into the undisturbed medium ahead of the arrest point, producing a unidirectional fan, which also explains why there is no backward radiation in the supershear case.

The v-shape observed in the sub-rayleigh case observed in figures ?? can be explained by the fact that the crack never outruns its radiated waves. This means that the stopping phase can propagate in all directions, unlike the supershear case (freundDynamicFractureMechanics1998). The slopes of the arms of the v are steeper than the one of the rupture front itself, which means that the stopping phases travel at faster speeds than the rupture. This means that the waves emitted by the arrest could for example travel at the highest medium speed (the p-wave or S0). In both the sub-rayleigh strain and strain rate as well as velocity plots, the backward-propagating wavefront is strongest at the snow bottom. This can be attributed to the reflection of the backward-propagating waves by the rigid bottom boundary (graffWaveMotionElastic1975).

The distinct rupture arrest signature in the sub-rayleigh versus supershear case has implications for DAS measurements because it could mean that the propagation regime of the crack could be determined from the crack arrest itself. The presence of a rupture arrest front also has implications for avalanche propagation. As gaumeInvestigatingReleaseFlow2019; gaumeDynamicAnticrackPropagation2018 state, the application of a load or disturbance somewhere on a slope can cause remote triggering of avalanches. Furthermore, burridgeAdmissibleSpeedsPlaneStrain1973; trotterTransitionSubRayleighAnticrack2022

have stated that one crack can trigger another (daughter crack nucleating from mother crack). This is important in relation to the rupture arrest because the wavefronts emitted by the arrest could trigger daughter cracks or even trigger avalanches remotely.

Dispersion and Broadening of Wavefronts

In the supershear strain and strain rate plots, the main wavefront weakens with increasing time (meaning the colors are less saturated). The same is true for the precursor in the sub-rayleigh strain and strain rate plots at the snow bottom. The weakening of the wavefront is observed clearly in the supershear case, because the crack travels faster than the guided waves (dunhamNearsourceGroundMotion2005). A Mach-front is created, where all the waves travel together in a "shock-front" (dunhamNearsourceGroundMotion2005), but once this front passes, there is attenuation of energy. In the sub-rayleigh case on the other hand, the guided waves are faster than the crack (kAkiRichards19801981), there is reflection of the waves between the top and bottom of the snow, which creates the oscillations which can be seen in figure ??.

In the sub-rayleigh case, the precursor wave also broadens and weakens with time (right column of figure ??). At the snow bottom, they are further away from their source (which is at crack depth). Because the S0 precursor is only emitted by the first and last source, its first arrival is strongest. After that, it loses energy because there is no source emitting this type of wave anymore, and the S0 precursor loses energy (kAkiRichards19801981), which can be seen by the colors fading. This wavefront does not broaden, it only starts fading with increasing time.

The broadening of the wavefront with increasing time can be explained by dispersion. By the reflections of the top and bottom snow boundary, the waves present start interfering with each other, which can lead to constructive or destructive interference (kAkiRichards19801981). This means that waves can start travelling at different phase velocities, because of the interference graffWaveMotionElastic1975. The phase and group velocities of waves are described as

$$v_{\text{ph}} = \frac{\omega}{k}, \quad v_g = \frac{d\omega}{dk}. \quad (5.1)$$

The interference leads to the broadening of the wavefront with increasing time, described by the envelope spreading law (kAkiRichards19801981)

$$\Delta x(t) = \Delta x_0 \sqrt{1 + \left(\frac{\frac{d^2\omega}{dk^2} t}{(\Delta x_0)^2} \right)^2}. \quad (5.2)$$

As time goes on, there are more reflections present, so there is more potential for interference and thus dispersion.

Amplitude decay and depth dependence of Strain and Strain Rate

The weakening of the wavefronts, especially observed in the supershear case in figure ?? and the precursor in ?? can be explained by depth decay and geometric spreading from the source. The crack tip acts as a moving two-dimensional point source, travelling along a line in a 2D elastic medium, the peak amplitude A of the radiated wavefield decays inversely with the square root of the radial distance r from the source plane (achenbachWavePropagationElastic2012). Wave components whose horizontal phase velocities are below the bulk wave speeds of the snow become evanescent in the vertical direction, decaying exponentially with depth from the crack plane (graffWaveMotionElastic1975). This explains why higher-frequency content is attenuated more rapidly between crack depth and snow bottom

The sub-Rayleigh strain and strain rate plots in figure ?? look very different for crack depth versus snow bottom. At crack depth, the main wavefront corresponding to the rupture is broad, while the S0 precursor is faint in energy and narrow. At the snow bottom on the other hand, the precursor is wider and decays in amplitude and the main wavefront is very narrow but high in energy. As discussed, the S0 precursor widens at crack depth because of geometric spreading (achenbachCHAPTER6WITHDRAWN1975). At crack depth, this precursor is narrow because it is impulsive and non-dispersive (roseWavesPlates2014). The main wavefront at crack depth is wide because is measured directly at the source, this means that the full field of the source is captured (roseWavesPlates2014), this includes the near-field of the source and the dispersive A0 coda emitted by the crack. At the snow bottom, the coda has dispersed, which leaves only the crack front which is resolved as a sharp line.

Magnitude of Wavefronts

In the supershear case (figure??), the two fronts merge into a single broad band because the rupture speed is within a few metres per second of the p-wave speed and the S0 speed. The strain and strain rate panels at both depths look essentially identical with no change in amplitude of strain or strain rate from crack depth to snow bottom. This contrasts sharply with the sub-Rayleigh case ??, where the amplitude drops significantly between the two depths.

A consequence of this difference in depth dependence of magnitude of strain is its ratio between the two regimes. In the supershear strain rate plots, the colorbar minimum and maximum is approximately one order of magnitude larger than in the sub-Rayleigh panels. Supershear ruptures are more efficient radiators of seismic energy to the free surface because the crack tip is co-located with the wave front rather than trailing behind it, concentrating energy into a Mach cone that hits the snow surface with maximum constructive interference (dunhamNearSourceGroundMotion2005; rosakisIntersonicShearCracks2002). In the sub-Rayleigh regime, the energy is spread over a broader spatial and temporal footprint because the rupture front and the precursor waves propagate at different speeds, reducing the peak strain rate at any point in space or time. This amplitude difference has a direct implication for field measurements: supershear avalanche releases would be substantially more detectable by DAS or than sub-Rayleigh events of the same moment release, and consequently, a DAS cable that can just barely detect a supershear event would be unable to detect a sub-Rayleigh event of equal size at the same distance. The same is implied for strain and strain rate at different depths in sub-Rayleigh propagation regimes. Because the magnitude of strain (rate) in figure ?? is one order lower at the snow bottom than at crack depth, the cable location relative to the nucleation is important. Since cables are usually installed before the first snow-fall (edmeFiberopticDetectionSnow2023), they would be located at the snow bottom rather than at the depth of the crack. This means that the sub-Rayleigh crack propagation might be too small in magnitude to be recorded by the DAS cable. Since bergfeldSignaturesSubRayleighSupershear2025; sironTheoreticalFrameworkDynamic2023; trotterTransitionSubRayleighAnticrack2022; gaumeDynamicAnticrackPro have shown that there can be a transition between sub-Rayleigh and supershear crack propagation, this means that the sub-Rayleigh onset of the crack might not be recorded in DAS because of its smaller magnitude.

5.1.2 Velocity space-time plots

Like in the strain and strain rate plots, the main wavefronts in the supershear and sub-Rayleigh case in figures ?? correspond in slope to the rupture speed of each case. Like in strain (rate), the velocity output is vastly different for sub-Rayleigh and supershear, which means a transition between propagation regimes can not only be observed in the strain output, but also in the velocity output.

Similarity to Strain and Strain Rate Plots

The sub-rayleigh horizontal strain and strain rate plots in figure ?? look very similar to the horizontal velocity plots in figure ?. The same is true for the supershear strain and strain rate in figure ??, which look similar to all but the vertical velocity at the snow bottom (top right in figure ??). This is intuitive because strain is the spatial derivative of displacement ($\frac{\partial u}{\partial x}$), strain rate is the temporal derivative of strain ($\frac{\partial^2 u}{\partial x \partial t}$). Velocity on the other hand is the spatial derivative of displacement ($\frac{\partial u}{\partial t}$), so all the values are derived from displacement. The strain and strain rate shown is in the horizontal direction, which explains why the vertical velocity looks different from the strain (rate) plots. In this section, only results which are specific to velocity will be discussed in detail, all other results are discussed and proved in detail in the strain and strain rate discussion.

Because the velocity plots are so similar to the strain and strain rate plots, the effects observed in both results are similar. Just like in strain and strain rate, there is broadening of the wavefront in the supershear case in figure ??, which can be explained by dispersion and envelope spreading, shown in the strain and strain rate discussion.

The amplitude decay of the main wavefront and the precursor wave discussed in the strain rate plots is present in the supershear case and the sub-rayleigh v_x at crack depth, it can also be explained by amplitude decay and depth dependence. The same is true for the width of the wavefronts in the sub-rayleigh case depending on location.

Two-wavefront structure in Sub-Rayleigh

The precursor wave which can be observed in the v_x plots in the sub-rayleigh case in figure ?? has a much faster speed than the rupture itself. This wave also only occurs in the sub-rayleigh case in the v_x plots. As for strain, simulations with larger source spacing (for example 10m spacing instead of 0.1m spacing, see appendix ??, figure ??) have shown that this precursor disappears if source spacing is increased. This leads to the same conclusion as discussed in the strain case that this is because of interference between the sources (madariagaDynamicsExpandingCircular1976).

The precursor wavefront is the symmetric Lamb wave mode (S0), whose propagation speed is limited by the plate wave speed:

$$c_p^{plate} = 2c_s \sqrt{1 - \frac{v_s^2}{v_p^2}} \quad (5.3)$$

(roseWavesPlates2014). When using the material parameters used in the simulation, this is just short of the p-wave speed of the slab. The speed of the S0 wave converges to the plate wave speed for low frequencies (long wavelength limit) (roseWavesPlates2014), which is why this wave is believed to be the S0 mode which is excited.

This two-wavefront structure with a fast S0/P-wave precursor followed by a slower crack-front signal is the signature of sub-Rayleigh rupture. The spatial separation between the two fronts grows linearly in time as

$$\Delta x(t) = (v_p - v_r) t, \quad (5.4)$$

where v_p is the P-wave speed and v_r is the rupture speed. This linear growth is directly measurable from the space-time plots in figure ?? and is consistent with the theoretical prediction for an expanding crack in a homogeneous elastic medium (freundDynamicFractureMechanics1998). The origin of the precursor is the redistribution of the stress field ahead of the crack tip, which propagates at the fastest available wave speed in the medium regardless of the crack tip speed (kAkiRichards19801981), so in this case, the p-wave or the S0 wave speed. Because the p-wave speed is very close to the S0 speed, this result is consistent with field observations of avalanches using seismic geophone arrays by

herwijkenSeismicSensorArray2011, a p-wave precursor to the main avalanche release was also observed. While herwijkenSeismicSensorArray2011 do not make note of the crack propagation regime, the p-wave or S0 precursor can be an important warning signal for sub-Rayleigh crack propagation leading to avalanche release.

The precursor is visible in sub-Rayleigh because the separation $\Delta x(t)$ grows fast enough to resolve the two fronts within the observation window. In the supershear case, the rupture front, S0 and the pressure wave all have velocities within a couple of metres per second of each other. Nevertheless, in the vertical velocity at the snow bottom in ??, two more distinct wavefronts can be seen, one which could be the S0 wave. This can be explained by the fact that v_y is the dominant component of the A0 mode (lambWavesElasticPlate1917), which at the snow bottom has sufficient geometric spreading to separate from the S0 precursor (roseWavesPlates2014). This means that the slower front is likely the A0/main wavefront from the crack, while the steeper one is the S0 mode. In all supershear velocity plots, at the end of the crack, there is a fanning out of waves which can be observed, which could be a combination of the stopping phases of the different wave types. This behaviour at arrest is consistent with the stopping-phase radiation described by madariagaHighfrequencyRadiationCrack1977, who showed that abrupt arrest of a crack generates a burst of high-frequency energy that propagates at the bulk wave speeds of the medium, so the p-wave speed, which is close to the S0 in this case, independent of the rupture speed before the arrest.

Lamb Wave Identification

The presence of waves with opposite polarity can be proven mathematically, the detailed derivation of this is shown in section ??.

The animations of the velocity wavefield in appendix ?? show the propagation intensity of the velocity wavefield. When comparing the horizontal velocity v_x in figure ?? with the wavefield intensity of the two lamb wave modes, for example in cartaLambWavesDiscrete2023, which can be found in appendix ??, it becomes evident that the main mode which can be observed is the antisymmetric (A0) mode. This is because at source depth, the velocity field has a lower intensity. Above and below the sources, there are narrower bands of low intensity (blue in figure ??, red in the reference figure ??) beside bands of higher intensity. This can be explained by the fundamental theory of lamb waves (lambWavesElasticPlate1917; roseWavesPlates2014). Because the moment tensor used here has a distinct out of plane component (negative m_{yy}), the primary mode which is excited is A0 (roseWavesPlates2014). As discussed, the S0 mode can also be seen, but is a precursor rather than the main wavefront and carries less energy. This precursor can also be seen in figure ??.

The dominance of A0 over S0 follows from source-mode coupling theory. For a moment tensor source applied normal to the slab midplane (so normal to the horizontal of the slab), the coupling coefficient to A0 scales as the bending moment integrated over the source region, while coupling to S0 scales as the in-plane force resultant (roseWavesPlates2014; achenbachWavePropagationElastic2012). Because the force applied has a negative y-component (M_{yy} of the moment tensor) and is normal to the slab axis, this maximizes A0 excitation. Because the force is only in the negative direction, it is asymmetric, so while A0 excitation is maximized, S0 coupling is minimized at low $f \cdot h$. This theoretical expectation is confirmed by the simulations: S0 appears primarily as a precursor in v_x (in-plane, extensional motion), consistent with the mode shapes described by lambWavesElasticPlate1917 and the coupling analysis of roseWavesPlates2014. This identification is strengthened by the animation stills in appendix ??, especially v_x in figure ?. The velocity field shows a polarity reversal around the crack, meaning that for example, the panel above the crack has positive polarity whereas the one below has negative polarity. Alongside of the comparison with figure ?? in appendix ??, this is proof for the A0 mode because S0 would have the same polarities on either side of the crack

(`lambWavesElasticPlate1917`).

5.1.3 Discussion of f-k-plot

Similarly to the results for the velocity and strain plots, the f-k-plots shown in section ?? show that the energy is concentrated around the p-wave speed (red line in figure ??) and the rupture speed (blue line in figure ??). In the supershear case, these two speeds are very close together, which leads to one energy band, whereas in the sub-Rayleigh case, they are more separated and two energy bands are observed. The S0 wave speed, as discussed previously, is also very close to the p-wave speed, which also has an effect on the concentration of energy in those areas.

In the sub-Rayleigh f-k plot in figure ??, the energy between the two dominant slopes (between the rupture speed and the P-wave speed lines) is not random noise. It corresponds to the dispersive A0 Lamb mode, which at different frequencies propagates at different phase velocities between v_r and v_p . The fan of diagonal stripes observed in the sub-Rayleigh v_y waterfall plots maps onto this wedge in f-k space: each stripe is a different frequency component of A0 arriving at a different time at a given location. This dispersive fan is a direct encoding of the A0 dispersion curve of the snow slab at $f \cdot h \approx 6 \text{ Hz m}$ (`roseWavesPlates2014`; `lambWavesElasticPlate1917`). In principle, the shape of this dispersive wedge could be inverted to recover the elastic properties of the slab (thickness, shear modulus) using the same surface-wave dispersion inversion methods applied in geophysical prospecting (`yilmazSeismicDataAnalysis2001`).

In the sub-Rayleigh case, the f-k plot in figure ?? shows a concentration of energy at the v_p of the slab and the S0 mode, as well as the rupture speed, consistent with the observations made in the velocity and strain plots.

This is also the case in the supershear f-k plots and the corresponding velocity and strain plots. Because the rupture speed is very close to the P-wave and S0 speed in the medium, energy is concentrated around these two. The narrowness of the energy concentration in the supershear case compared to sub-Rayleigh reflects the absence of the dispersive two-front fan: the Mach cone geometry radiates energy preferentially at the cone angle $\alpha_{Mach} = \sin^{-1}(c_s/v_r)$ (`dunhamNearSourceGroundMotion2005`; `rosakisInterSonicShearCracks2002`), which at $v_r \approx v_p$ collapses to a nearly flat front, concentrating all radiated energy into a single, tightly bounded slope in f-k space.

The strong contrast in how the energy is concentrated between the sub-Rayleigh and supershear case in ?? can be explained by the number of locations the energy is concentrated. As described above, the supershear regime has energy at essentially one slope, whereas the sub-Rayleigh case fills a wedge between two slopes. This explains why the energy bands in the supershear case appear much narrower and more concentrated compared to sub-Rayleigh.

The observation that at higher frequencies in figure ??, the lines of the energy present do not match as well anymore with the concentration of energy, can be explained by numerical dispersion. As `komatitschIntroductionSpectralElement1999` describe, as frequency increases, the wavelength decreases. This means that higher frequencies are not resolved as well by a spatial grid with a certain spacing `yilmazSeismicDataAnalysis2001`, this starts bending the energy in the f-k plot. This could explain the mismatch between the physical velocity lines and the concentrations of energy.

The horizontal striping visible in the sub-Rayleigh f-k plot at constant frequency (for example radiating from the v_p -line) is a temporal aliasing artefact arising from the finite output sampling interval of the simulation, not a physical feature. This can be distinguished from the physical dispersion fan by the fact that the stripes are perfectly horizontal (constant frequency, for all wavenumbers), whereas physical A0 dispersion produces curved or oblique features in f-k space.

5.1.4 Implications for DAS measurements & Synthesis of Results

The results in section ?? and their discussion have shown that there are multiple implications for the measurement and interpretation of DAS data, even in a simple physics case.

In all sub-rayleigh results, the presence of a precursor wavefront, which is the S0 wave, is shown. This is consistent with field observations made by herwijnenSeismicSensorArray2011, but the precursor is also weaker in amplitude compared to the main wavefront and decays in amplitude with time. This is important because it shows that the precursor wave might not be a reliable "early warning" because it might not be detected on account of its low magnitude. The main wavefront travels at crack propagation speed and does not undergo significant amplitude decay.

In all supershear results, there is only one wavefront because the S0 and p-waves have a similar speed to the rupture. The precursor can be identified by looking at the vertical velocity at the snow bottom as well as the rupture arrest. In the supershear case, the wavefront is dispersive and decays in amplitude with depth, it also broadens with increasing propagation time. This is because the rupture and the precursors travel together as a Mach front, which is attenuated (dunhamNearSourceGroundMotion2005). The broadening can be explained by dispersion, so the change in phase velocities caused by reflections and interference (graffWaveMotionElastic1975).

The presence of the arrest phase in both the supershear and sub-rayleigh propagation regime and their different manifestations could be important in further distinguishing propagation regimes. While sub-rayleigh has a v-shaped arrest, supershear propagation has a fan-like arrest front that only propagates in the positive direction. This could help identify the propagation regime of a crack in snow, which in turn is important for early warning. The faster a crack propagates, the faster the critical area needed for release is reached and the avalanche will be released (bergfeldSignaturesSubRayleighSupershear2025; trotterTransitionSubRayleighAnticrack2022).

The supershear velocities and strain (rate) has a higher amplitude than their sub-rayleigh counterparts. This means that supershear ruptures can be detected more easily in DAS measurements because they radiate more energy. In the sub-rayleigh case, there is also significant difference in the amplitudes of strain, strain rate and velocity between the crack depth and the bottom of the snow, the bottom of the snow having tenfold lower amplitudes. This is crucial for DAS field measurements because the cable is usually located at the bottom of the snow. The implications for field measurements are that the sub-rayleigh propagation preceding the supershear transition (trotterTransitionSubRayleighAnticrack2022) might not be seen in DAS data because it has too low amplitude. This is important when interpreting field data because false conclusions of pure supershear rupture might be drawn from this.

The gauge length used to simulate DAS measurements here is 10.4m, like in turquetAutomatedSnowAvalanche2024. This has implications for how the results look because some wavetypes might be smoothed over on account of their wavelength. The S0 mode which has a speed around $v_p \approx 109\text{m/s}$ and A0 is similar to the rupture speed, in sub-rayleigh this is $v_r \approx 40\text{m/s}$. Calculating the wavelength from speed according to $\lambda = \frac{v}{f_0}$ (kAkiRichards19801981) means that the S0 mode has a larger wavelength. This means the larger the DAS gauge length, the more the A0 mode is suppressed because one DAS gauge length spans more A0 wavelengths. To make sure the gauge length of 10.4m does not lead to misrepresentation of the results, data for a 1m DAS gauge length were simulated, these can be found in appendix ?. The supershear case shown in figure ?? at 1m gauge length looks almost identical to the strain at 10.4m gauge length in figure ?. The sub-rayleigh strain looks slightly different at a smaller gauge length (?), compared to the turquetAutomatedSnowAvalanche2024 gauge length in the results figure ?. This comparison shows that the observations made in the sections above are valid for small as well as large gauge lengths. This is important for because it shows that crack propagation looks similar no matter the gauge length of the DAS cable used.

5.2 Material Point Method Guided Simulations

In the following sections, results from the material point method guided simulations will be discussed. Because many phenomena which are observed in the simple model are also observed in the MPM model, these will not be discussed in detail.

5.2.1 Strain and Strain rate space-time plots

Propagation speeds & Wavefront geometry

The v-shaped wavefronts which can be observed in figure ?? are likely emitted by the crack itself. The slope of these two arms in both sides of propagation does not change from the vertex, which is the crack onset, to the end of the domain. This indicates a constant propagation speed, which means unlike simulations and experiments by sironTheoreticalFrameworkDynamic2023; bergfeldSignaturesSubRayleighSupershe there is no transition in crack propagation speed. When taking the slope of the positive strain wavefront which arrives first in figure ??, this results in: $v_{prop} = \frac{125-75}{1.2-0.5} = 57.14 \text{ m/s}$. Considering the medium velocities used (shown in table ?? but also in the f-k plot ??), this is sub-rayleigh propagation in both the weak layer but also the snow slab.

The slope of the backward propagating arm in figure ?? is calculated according to the values in table ??, the same is true for the second wavefront which is observed as well as the forward-propagating wavefront before the crack arrest.

Wave type	Slope formula	Calculated speed
Between crack onset and arrest	$v_{prop} = \frac{125-75}{1.2-0.5}$	57 m/s
Backward propagating	$v_{prop} = \frac{75-0}{1-1.6}$	-125 m/s
Second wavefront	$v_{prop} = \frac{175-150}{4-3}$	25 m/s
Forward propagating after crack arrest	$v_{prop} = \frac{200-125}{1.8-1.2}$	125 m/s

Table 5.2: Speeds calculated from slopes in figure ??.

From the speeds calculated from the mpm strain results shown in table ?? it becomes more clear which wave types are observed.

The crack propagation speed of 57 m/s, which is in the sub-rayleigh regime, is consistent with the conditions needed for transition between the sub-rayleigh and supershear regimes. trotterTransitionSubRayleighAnticrack2022 state that on flat terrain, the normal stress at the crack tip is higher than the shear stress in the same location. This drives anticrack propagation, which is in the sub-rayleigh regime (gaumeDynamicAnticrackPropagation2018 trotterTransitionSubRayleighAnticrack2022). For a transition to the supershear regime, a slope would be needed which drives a transition from a mixed-mode anticrack propagation to pure mode II crack propagation (trotterTransitionSubRayleighAnticrack2022). The moment tensor components which were used as STFs shown in figure ?? all show negative amplitudes. For a mode II crack, which is in-plane shear, the component shown in ?? would have to have an alternating sign (red and blue), alongside m11 and m22 shown in ?? and ?? very small or zero. Since all three components have a strong negative amplitude, a mixed-mode anticrack is more likely, which would lead to sub-rayleigh crack propagation.

The speeds of the forward and backwards-propagating waves emitted by the crack is the same in the positive as well as the negative direction. This means that this type of wave is emitted on both sides of the crack. Its speed is between the p-wave speed of the weak layer and the p-wave speed of the snow slab (table ??). This is consistent with the results discussed in the simple model section ??, where a sub-rayleigh crack propagation regime emits an S0 precursor, which is also consistent with field observations of herwijnenSeismicSensorArray2011.

The third wavefront which is observed in figure ??, has a much lower propagation speed than all other wavefronts. This wavefront also broadens with time and the waves curve downward, towards the origin of the coordinate system. This wavefront could be the dispersive A0 Lamb wave mode, because at low frequency thickness product, and at the dominant frequencies present in the model, has a group velocity which is significantly below the S-wave speed of the slab (roseWavesPlates2014).

Wavefield Structure & Quasi-Static Strain field

The most striking feature identified in the strain and strain rate plots is the thick bands of positive and negative polarity coming from the crack onset. In the positive direction of propagation, the first arrival is a positive strain or strain rate, followed by a negative (blue) band of negative strain (rate). In the negative direction of propagation, this order is reversed, the negative polarity arrives first. This band changes when the crack has stopped: at $x=125\text{m}$, the positive wavefront suddenly jumps to be wider. In the region of the crack, so from 75m to 125m, it was a narrow wedge whereas the blue region was very broad.

The positive wedge of energy is the quasi-static strain field, as described by FreundDynamicFractureMechanics1998. It builds up progressively between $x = 75\text{ m}$ and $x = 125\text{ m}$ as the crack propagates, then transitions abruptly to a broad negative region (blue) after the crack arrests at $x = 125\text{ m}$. This red wedge is narrow at the crack onset ($t \approx 0.5\text{ s}$, $x = 75\text{ m}$) and widens to its maximum at the crack arrest ($t \approx 1.2\text{ s}$, $x = 125\text{ m}$), tracing the spatial footprint of the growing crack. The abrupt widening of the positive front at $x = 125\text{ m}$ marks the moment of arrest, where the accumulated quasi-static compression is released as the crack stops, and the strain field relaxes into its post-failure configuration. This arises because the moment tensor STF has a non-zero DC-component and there is permanent deformation, which is described in section ??. This means that the weak layer undergoes permanent deformation after the crack has passed. The deformation resulting is collapse, which is consistent with an anticrack as described by HeierliAnticrackNucleationTriggering2008; GaumeDynamicAnticrackPropagation2018; TrotterTransitionSubRayleighAnticrack2022, which can trigger avalanches. This is also supported by the moment tensor components shown in figure ??, where all components are predominantly negative. The region before crack propagation (white) can be clearly distinguished from the region where the crack is propagating or has propagated, which have a negative value. This means that there are no restoring forces after the crack has propagated, which means that the deformation stays in place. This is further supported by the cumulative integral of all moment tensor components shown in figure ??. All three cumulative integrals approach a negative value, which means that the displacement caused by them is also negative. In other words, because of the asymmetric shape of the moment tensor source time functions used in the MPM simulation, which do not integrate to 0 over time, there is a net displacement. Since all components result in a negative displacement, this results in an anticrack, or weak layer collapse.

The quasi-static wavefield is contained within a v-shaped wavefront originating from the crack onset at $x=75\text{m}$. It shows high amplitude lines of positive and negative strain which are alternating. In the positive direction ($x > 75\text{ m}$), the first dynamic arrival is positive strain or strain rate (red) followed immediately by negative strain (rate), consistent with the extension-first, compression-second polarity sequence of P-wave or S0 Lamb mode radiation (KarkiRichards19801981). In the negative direction ($x < 75\text{ m}$), this order is reversed because the receiver is on the opposite side of the source: the compressive push from the crack initiation reaches a point to the left as a compression-first arrival. The slope of the forward-propagating arm gives the crack propagation speed ($v_r \approx 57\text{ m/s}$, in the region of the crack, and $v \approx 125\text{ m/s}$ after the crack has propagated, seen in table ??). The slope of the backward-propagating arm gives the P-wave or S0 speed ($v \approx 125\text{ m/s}$), consistent with the precursor identified in the simple model results and with field observations of HerwijnenSeismicSensorArray2011.

The second dispersive wave train observed in the crack depth panels on the left side of figure ?? is the third feature which was identified. This front manifests as lower amplitude lines propagating from the crack onset. They alternate in polarity and have a steeper slope, which means lower apparent velocity (see table ??). These are the same alternating polarity bands identified in the simple model as the dispersive A0 Lamb wave mode. Their identification as A0 is confirmed in the velocity discussion (section ??), where the polarity reversal across the slab midplane is directly visible. Unlike the dynamic V-shaped wavefronts, these secondary bands broaden with increasing time and are absent from the snow bottom panel (right column), consistent with the evanescent depth decay of the A0 mode at this frequency-thickness product (roseWavesPlates2014), which was also identified in the simple model discussion.

5.2.2 Curving and broadening of wavefronts

The secondary wavefronts observed in the crack depth plots, which are shown in the left column of figure ??, spread out with time. This spreading, as in the simple model discussion, can be explained by dispersion. This is consistent with the dispersive A0 lamb wave mode discussed in section ??. The same dispersion mechanism applies here: reflections between the slab boundaries produce interference between frequency components travelling at different phase velocities, broadening and curving the wave train envelope with time (graffWaveMotionElastic1975).

5.2.3 Velocity space-time plots

Similarity to Strain (Rate) plots

As discussed in section ??, velocity, strain and strain rate are all derivatives of displacement. Since the horizontal component of strain and strain rate is shown and discussed for the MPM model, the velocity discussion section will only go in detail about phenomena which can be seen better or exclusively in the velocity plots.

Propagation speeds

The propagation speeds of the wave types observed in the strain and strain rate plots shown in table ?? are consistent with those observed in the velocity plots in figure ??. The very saturated wavefronts in the horizontal component correspond have the same slope as the S0/P-wave precursor observed already in the strain and strain rate plots. The same is true for the very faint wavefront observed in the v_y plot at crack depth (top left panel of figure ??), which also has the same slope as the S0 wave. This means that the precursor can also be observed in the velocity data.

The sub-rayleigh crack propagation speed which was calculated from the strain and strain rate plots is the same speed which can be calculated from the slope of the line connecting the crack onset and arrest in figure ??.

Quasi-static offset

The quasi-static offset is also present in the velocity plots, mainly the horizontal velocity plots in the bottom row of figure ??. Contrary to the strain and strain rate plots, the first arrival in the positive direction of propagation has negative velocity, followed by positive velocity, but it is also reversed in order in negative direction of propagation. This polarity reversal between velocity and strain for the same compressive anticrack source is expected, because a compressive source produces extensional strain ahead of the source but a negative initial velocity as material is displaced away from the crack before elastic restoring forces act (kAkiRichards19801981). The quasi-static offset can also be seen

in the animation of the velocity wavefield in appendix ??, figure ??, where the quasi-static offset manifests as broad areas of negative velocity followed by positive velocity. This is the kinematic signature of anticrack collapse: the material above the weak layer is permanently displaced in the negative direction as the weak layer fails under pressure. This is consistent with the anticrack model described by McClungFractureMechanicalModels1981, and the negative DC component of the moment tensor shown in ??.

Lamb wave identification

Like the strain and strain rate plots, all velocity plots except v_x at the snow bottom, show secondary wavefronts of alternating polarity. This can also be seen very well in the animation of the vertical component of the wavefield, a still of which is shown in appendix ??, figure ??, where behind the crack location bands of red and blue (positive and negative vertical velocity) alternate. The same can be seen in the v_x still, but here the bands are much more broad.

In both animation stills ?? and ??, the crack is located at 100m. In the animation of the vertical velocity component, the velocities not only alternate in polarity, but they also switch polarity on either side of the crack. The crack depth, which is in the middle of the weak layer for example has positive polarity above it and negative polarity below it, which is characteristic for the asymmetric Lamb wave mode A0. The detailed derivation of the Lamb wave equations can be found in section ?. This polarity pattern is also consistent with figure ?? in appendix ??, showing the propagation intensities of the symmetric and antisymmetric Lamb wave modes. The A0 Lamb wave mode dominates the MPM wavefield for the same reason as in the simple model: the compressive normal component $m_{22} < 0$ of the moment tensor (shown in figure ??) is oriented normal to the slab and maximally couples to the A0 flexural mode at the low frequency-thickness product of this geometry (RoseWavesPlates2014). The MPM results therefore confirm the Lamb wave mode identification made in the simple model.

Rupture arrest structures

In the vertical velocity plots in figure ??, two secondary wavefronts can be observed. One coming from the crack onset, which is in the foreground and can be seen clearly, and the other coming from the crack arrest point at $x=125\text{m}$. This structure also has a v-shaped structure, which is consistent with the rupture arrest structure observed in the simple sub-Rayleigh plots.

Again, the crack never outruns the waves which are radiated from it, which means that the stopping phase can radiate in all directions (FreundDynamicFractureMechanics1998). The crack arrest at $x=125\text{m}$ produces a v-shaped stopping-phase signature consistent with sub-Rayleigh arrest, analogous to the arrest structure identified in the simple model (section ?). Both arms of the V propagate at the P-wave and S0 speed of the slab (125 m/s , measured in table ??), confirming that the stopping phase radiates in all directions at the bulk wave speed, independent of the pre-arrest rupture speed (MadariagaHighfrequencyRadiationCrack1977).

5.2.4 F-k plot

The f-k plot confirms the structures which have been observed in the previous sections. The f-k spectrum of the MPM-guided simulation (figure ??) shows two pairs of symmetric energy bands, one on each side of the wavenumber axis.

The steep inner bands in figure ??, aligned with the P-wave speeds of the slab and weak layer ($v_p \approx 127\text{ m/s}$ and $v_p^{\text{wl}} \approx 116\text{ m/s}$, shown by the red and white lines), correspond to the S0/P-wave precursor radiating from the crack initiation and arrest points at the fastest available wave speed. This is not just because the p-wave speeds in the slab and in the weak layer are close in value, but also because

the crack initiates in the weak layer and radiates into the slab. These bands appear on both the positive and negative wavenumber sides because the precursor radiates in both directions from the initiation point at $x = 75$ m, consistent with the forward and backward propagating arms of the v after the crack locations, measured at approximately 125 m/s in table ?? and with the precursor identified in the simple model sub-Rayleigh results. Because the slab and weak layer P-wave speeds are nearly identical in the MPM material (table ??), the combined energy band appears broader than either line alone. This is consistent with the same precursor observed in field measurements by herwijnenSeismicSensorArray2011, where a fast P/S0-wave arrival preceded the main crack signal.

The shallower outer bands, with a slope corresponding to $v_{rupture} \approx 57$ m/s from table ??, are the crack propagation front itself. The crack radiates energy both forward (in the direction of propagation) and backward (as each source fires and sends energy behind the advancing tip), producing bands on both the positive and negative wavenumber sides. The positive side is more concentrated and stronger in amplitude, reflecting the net forward propagation of the crack from $x = 75$ m to $x = 125$ m. The apparent slope of these bands gives a rupture speed consistent with the direct measurement from the strain panels, providing an independent f-k confirmation of the sub-Rayleigh propagation regime.

The dispersive A0 Lamb mode, which in the simple model produced distinct features between the rupture speed and P-wave speed lines in figure ??, does not appear here. This is consistent with the shorter crack length in the MPM model (40 m versus 240 m in the simple model): a shorter source array produces less coherent beaming of A0 energy into a concentrated f-k ridge, spreading it instead into a diffuse background (roseWavesPlates2014). This background could be contained in the concentration of energy around the two p-wave speeds.

Unlike in the simple case f-k-plot, the spectrum shown in figure ?? is symmetric around positive and negative wavenumbers. Unlike in the simple model where the long crack (240 m) produced strong directivity asymmetry between positive and negative k , the shorter MPM crack (40 m) has weaker directivity because the directivity effect scales with crack length relative to wavelength (madariagaDynamicsExpandingCircular2001).

5.2.5 Implications for DAS measurements

The MPM-guided results have several implications for the interpretation and measurement of field DAS data. They extend and support the conclusions drawn from the simple model results.

Validation of simple model signatures in the five-layer geometry

Before comparing the MPM and simple model results directly, it is important to establish that the wave signatures identified in the two-layer simple model are not an artefact of the simplified geometry. To test this, the same sub-Rayleigh and supershear prescribed rupture speeds used in the simple model were applied in the five-layer MPM geometry shown in figure ?? using Ricker source time functions, as described in section ?. The resulting strain rate and velocity panels (appendix ??) show the same two-wavefront structure, the same dispersive A0 wave train, the same rupture arrest signatures, and the same amplitude depth dependence as the two-layer simple model results. This means that the five-layer geometry does not introduce additional wave types or change in the DAS signatures identified in section ?.

This confirms that, firstly, the physical wave signatures discussed throughout this thesis are robust to the choice of model geometry and are not specific to the simplified two-layer domain. Second, any differences observed between the simple model and the MPM-guided results can be attributed to the MPM source time functions and the more realistic physics rather than to the geometry change. This isolates the effect of the source physics on the DAS output. This separation is important for interpreting the additional features observed in the MPM results, specifically the quasi-static strain field and

the different amplitude structure, as consequences of the MPM source rather than consequences of the five-layer geometry. This is why the two results can be compared to each other in the following section even though they have a different domain geometry.

Sub-Rayleigh propagation effects

The crack propagation speed measured from the MPM results ($v_r \approx 57$ m/s, table ??) is sub-Rayleigh for both the slab and weak layer material parameters. This is consistent with studies by trotterTransitionSubRayleighAnticrack2022; sironTheoreticalFrameworkDynamic2023; bergfeldSignaturesSubRayleigh2022 which show that transition between sub-rayleigh and supershear crack propagation occurs at high slope angles and the normal to shear stress ratio at the crack tip. Because of the moment tensor input, which shows constant mixed-mode anticracks and no indication of pure mode II, which would be indicative of the transition (trotterTransitionSubRayleighAnticrack2022), a sub-rayleigh regime is not only supported by the results, but also by the input. The DAS signatures observed in the MPM strain rate panels (figure ??) and velocity panels are therefore those of a sub-Rayleigh crack at physically realistic material parameters and STFs, rather than the simplified speeds and Ricker wavelets used in the simple model. The two-wavefront structure, which is a fast S0/P precursor and a slower crack front, is present in the MPM results as well as in the simple model, confirming that this signature is not an artefact of the prescribed source geometry but reflects genuine wave physics of sub-Rayleigh crack propagation in snow. This is also confirmed by field data by herwijnenSeismicSensorArray2011.

5.3 Comparison between pure SE and MPM

5.3.1 Propagation speeds & wavefronts observed

two wavefront structure might not be observed because speeds are much closer (propagation and p-wave speed)

5.3.2 Precursor & Rupture Signature

In the sub-rayleigh case in section ??, a precursor was identified, which travels in positive direction of propagation at the S0 wave speed in all plots. In some of the sub-rayleigh plots in the simple case, for example strain and strain rate at the snow bottom in figure ??, this wavefront can also be observed propagating in the negative direction. In the MPM-guided simulations, a wavefront with speeds close to S0 and the p-wave speed was observed propagating in the positive as well as the negative direction of propagation in all plots. [explain why](#)

While this precursor is weak in amplitude in the simple case plots, it can be seen clearly in the MPM-guided results. [why?](#) This comparison is important for the interpretation of DAS field data, because the precursor might not always look the same. The different amplitudes but also polarity structure of the precursor S0 wave leads to the conclusion that it can be best identified via its speed.

5.3.3 Quasi-Static Wavefield

In the MPM-guided simulations, discussed in section ??, the quasi-static displacement field can be observed clearly as broad red and blue wavefronts. Such structures cannot be observed in the sub-rayleigh simple case nor the supershear simple case. The reason for this is the source wavelet used. While the ricker wavelet, which was used in the simple case, is symmetric (kAkiRichards19801981), the MPM STFS are not symmetric. This means that their cumulative integral, unlike that of the ricker, is nonzero (kaganPointSourcesElastic1987; zhuNoteDynamicStatic2002; freundDynamicFractureMechanics1998), which leads to a net deformation, which is negative in this case.

5.3.4 Wave types observed

5.3.5 Crack arrest signatures

crack arrest much broader in mpm case

- Compare f_k plots and slopes measured: which propagation regime is observed in mpm?
- Quasi-static offset comparison: why absent in simple model and why in MPM, what implications does this have
- Comparison between wavefront structures: eg precursor
- Comparison between types of waves present: which lamb wave modes and why
- Crack arrest signature comparisons
- Why absence / presence of dispersion and spreading?

Outlook

The results of this thesis establish a baseline for DAS signatures of sub-Rayleigh and supershear crack propagation in a simplified geometry; the following extensions would improve the physical realism and practical applicability of the model. Based on the results and methods presented in this thesis, there are many open lines of inquiry. Because the model described in the simple case is a basic two layer model, a next step could be to add multiple layers, especially because cracks in snow usually propagate in a weak layer (schweizerSnowAvalancheFormation2003). Here, there is also possibility to run the simulations described in this thesis for different snow parameters (density, v_p and v_s , elastic modulus), because barteltPhysicalSNOWPACKModel2002; mulakNumericalInvestigationMixedmode2019; bobillierMicromechanicalModelingSnow2020 have all shown that snow properties not only change over time, but also have an influence on crack propagation. This would be useful to better understand DAS data from different times of the year or day, when there is the possibility of snow properties having changed.

There is also the possibility of adding topography or a slope angle to the simulation, because as indicated by anceySnowAvalanches2001; trotterTransitionSubRayleighAnticrack2022, avalanches are gravity driven and the crack propagation regimes can change greatly with the presence of topography.

Because of the possibility for transition between the different propagation regimes, meaning that the propagation speed changes, as described by trotterTransitionSubRayleighAnticrack2022, which is not only driven by topography and slope angle. Another next step would be to have source propagation accelerate with increasing propagation distance, instead of having a constant propagation speed, which is not very plausible on a mountain slope.

Furthermore, the crack modeled in this thesis is linear and at constant depth. MPM simulations of weak layer cracking and collapse have shown that this is not the case (trotterTransitionSubRayleighAnticrack2022), so a next step would be to make the crack propagation modeled by seismic sources more physical. This could be done by having "jumps" of the crack into different depths of the snowpack or having multiple lines of sources firing at the same time to represent multiple smaller cracks.

The last step in making the model described here physically accurate is to implement the 2D model described here in 3D, which would help to understand slope DAS data even better. Such a 3D model could represent data recorded further away from the crack, in this thesis, the 2D geometry constrains the cable to be parallel and or co-planar to the crack, whereas in the field, cables may be oriented at arbitrary angles to the crack propagation direction. It would also be very interesting to compare different cable orientations relative to the crack to see if the wavefield data recorded changes.

Conclusion

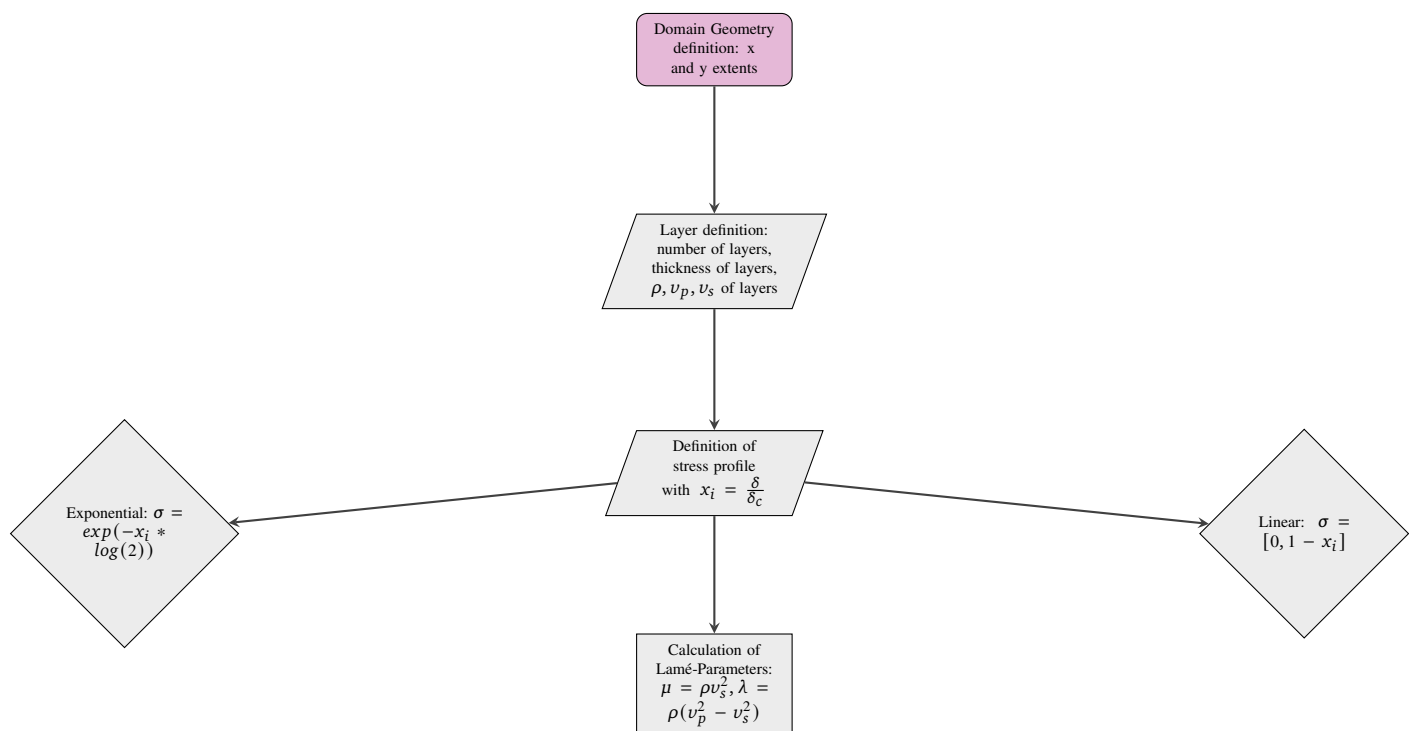
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Replace this placeholder with your conclusion.

Code Repository

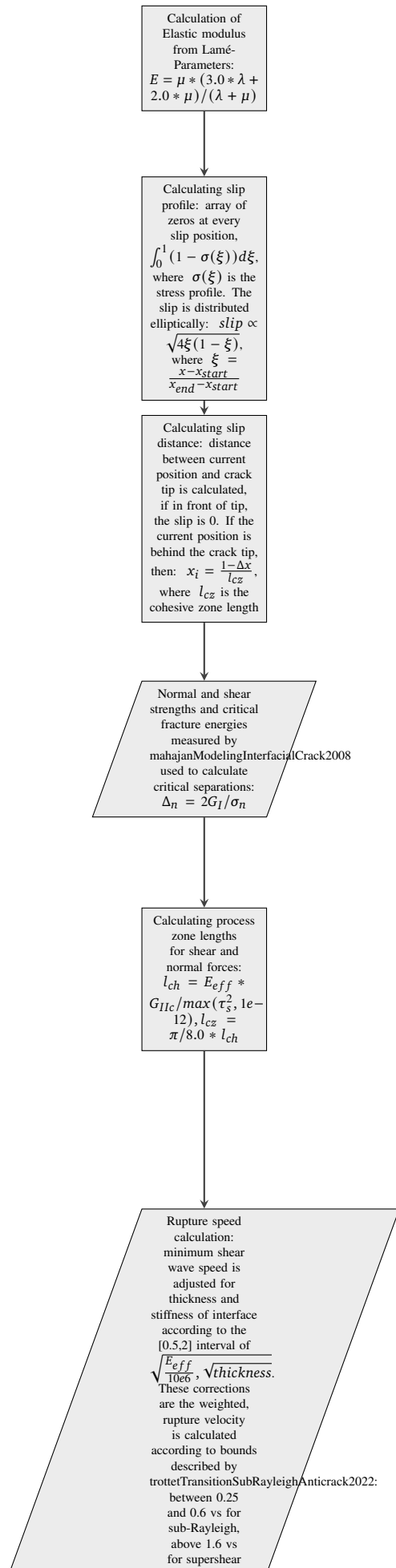
The complete code of all models and plots described can be found on github: <https://github.com/s-bachmann/Master-Thesis>. Meeting slides contains slides used for intermittent meetings with supervisors, Report contains all the $\text{\LaTeX}$ documents and auxillary files used to write this thesis, and Test Simulations contains all the simulation and plotting code.

Code Workflow fo CZM Model

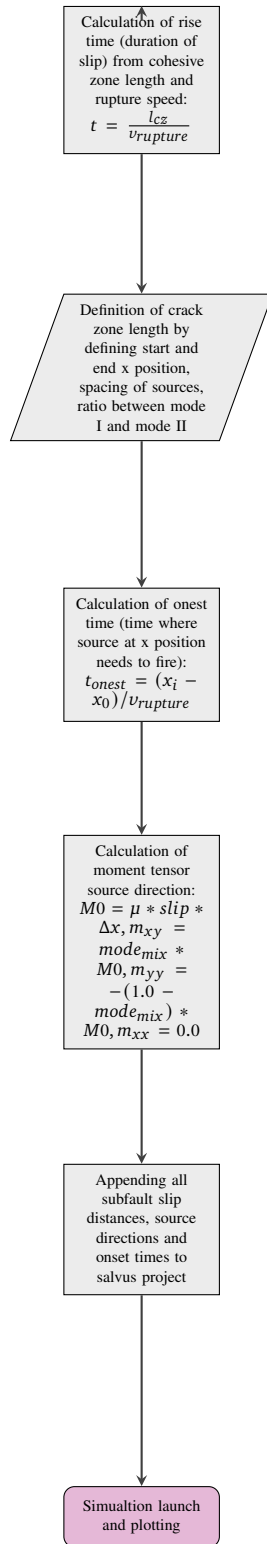
Part 1: Domain and Stress Profile Setup



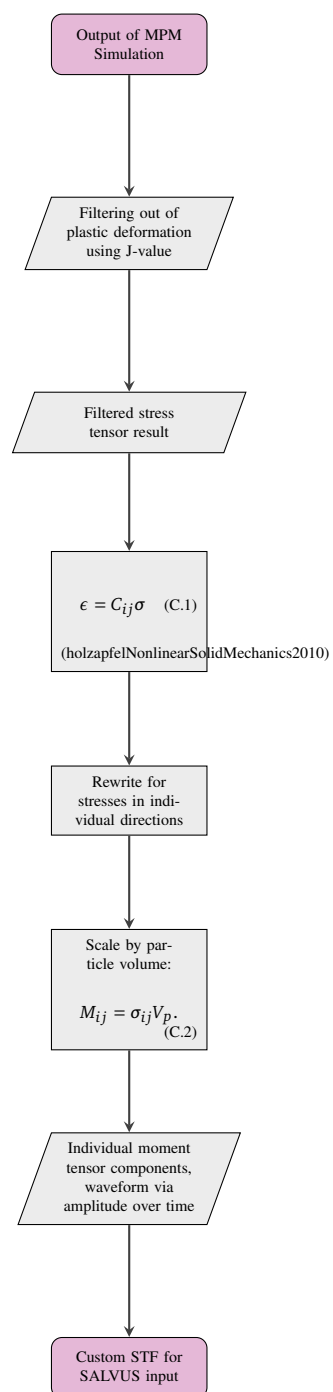
Part 2: Material Properties and Process Zone Calculation



Part 3: Rupture Propagation and Source Definition



MPM conversion workflow



Parameters used in Models

Parameter	Typical value / source
σ_n	0.00184 MPa from mahajanModelingInterfacialCrack2008
τ_s	0.004 MPa from mahajanModelingInterfacialCrack2008
G_I, G_{II}	0.35, 0.50 J/m^2 from mahajanModelingInterfacialCrack2008
ν_s, ν_p, ρ	67.9366, 109.4 m/s, 250 kg/m ³ from trotterTransitionSubRayleighAnticrack2022 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)

Parameter	Value
from trotterTransitionSubRayleighAnticrack2022	
θ	45°
$\rho_{\text{slab}}, \rho_{\text{bed}}, \rho_{\text{wl}}$	250.0, 300.0, 100.0 kg/m ³
$E_{\text{slab}}, E_{\text{bed}}, E_{\text{wl}}$	$3 \times 10^6, 4 \times 10^6, 1 \times 10^6$ Pa
ν	0.3 (Poisson's ratio)
Δx	0.01 m
Δt_{frame}	1/500 s
F	Deformation gradient tensor (F_{00} in xx , F_{11} in yy)
T	Stress tensor (T_{11} in xx , T_{22} in yy)
P	Particle position (x, y, z)
LogJP	Cumulative plastic damage (> 0)

Wider Source Spacing Plots

The following figure shows particle velocity (v_x and v_y) for a sub-rayleigh case with 10m source spacing. All other parameters apart from the spacing are the same as for the results shown in section ??.

The results for strain show a similar effect:

Sub-Rayleigh: particle velocity from d/dt displacement at $y = 2.625$ (Crack Depth) and $y = 2.25$ (Snow Bottom)

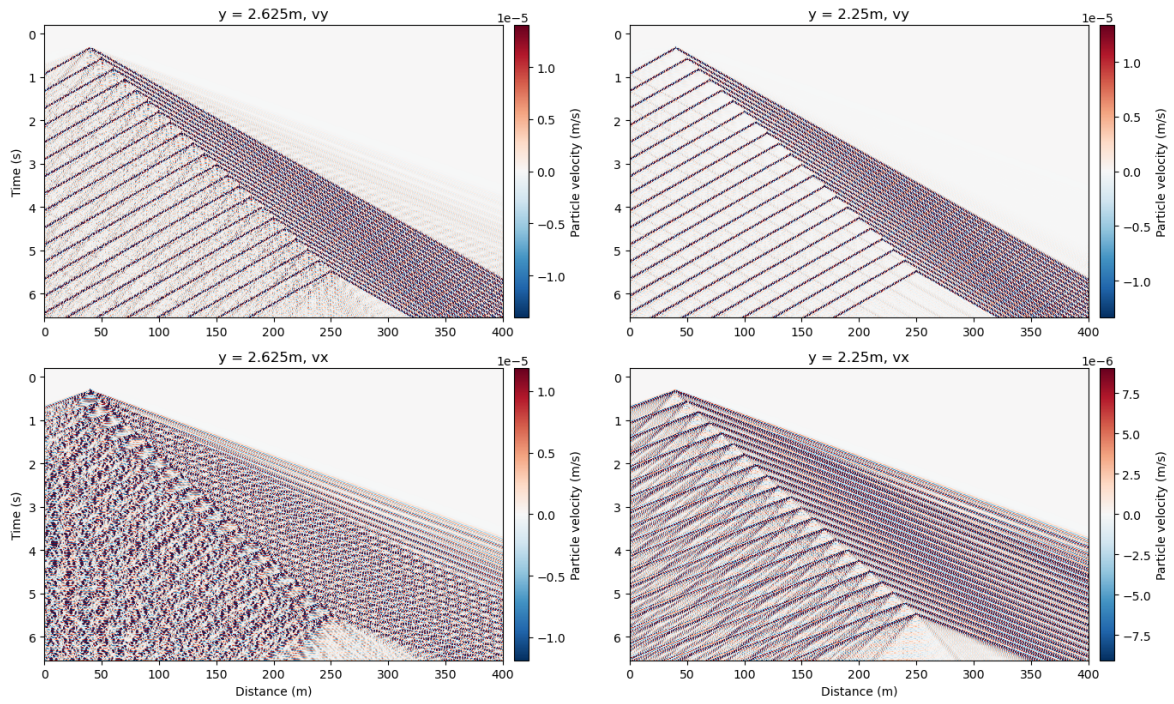


Figure E.1: Sub-rayleigh Particle velocity. Vertical velocity in top row, horizontal velocity in bottom row. Crack depth in left column, snow bottom in right column. Colorbar shows velocity (m/s). Source spacing is 10m.

Sub-Rayleigh: Pyber DAS Simulation (Gaussian, width=4.0m)
at $y = 2.625$ m (Crack Depth) and $y = 2.25$ m (Snow Bottom)

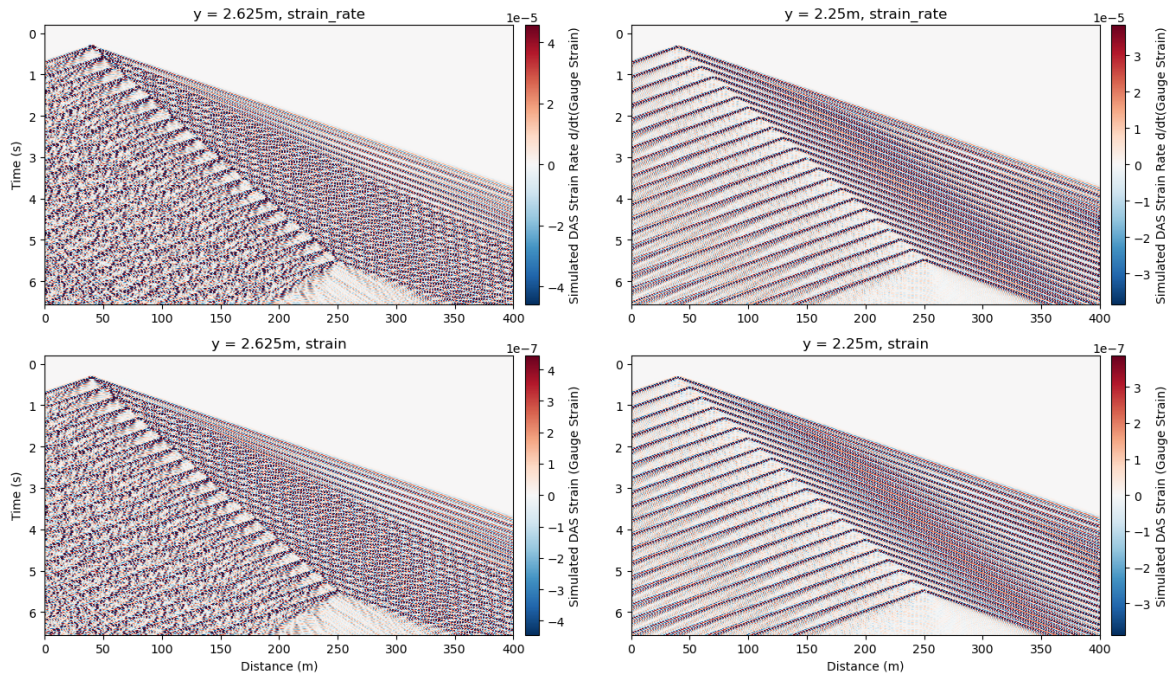


Figure E.2: Sub-rayleigh strain and strain rate. Crack depth in left column, snow bottom in right column. Averaging over 4m gauge length, strain rate in top row, strain in bottom row. Source spacing is 10m.

Animations of wave propagation fields

F.0.1 Simple model

The following two animation snapshots show the velocity field for the sub-rayleigh case in the simple model.

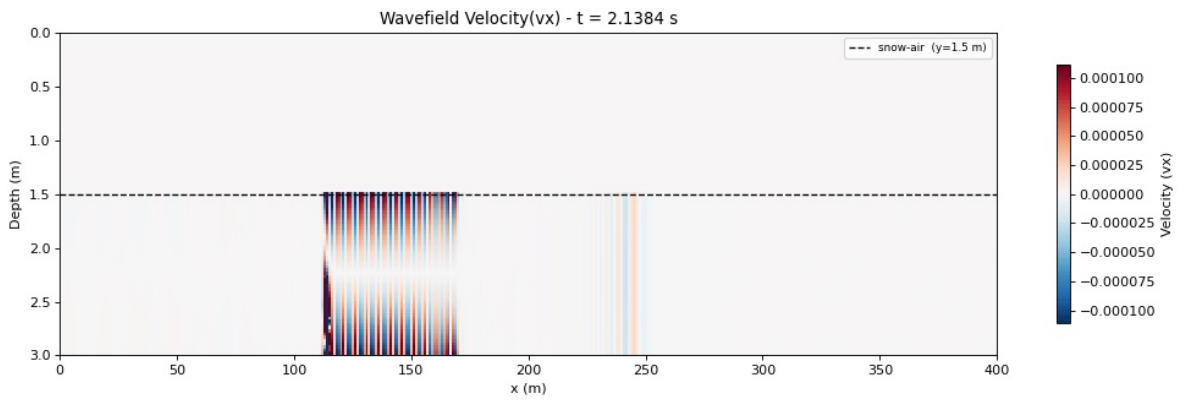


Figure F.1: Snapshot of the horizontal wavefield velocity component (v_x) evaluated at the crack location ($y = 2.625$ m).

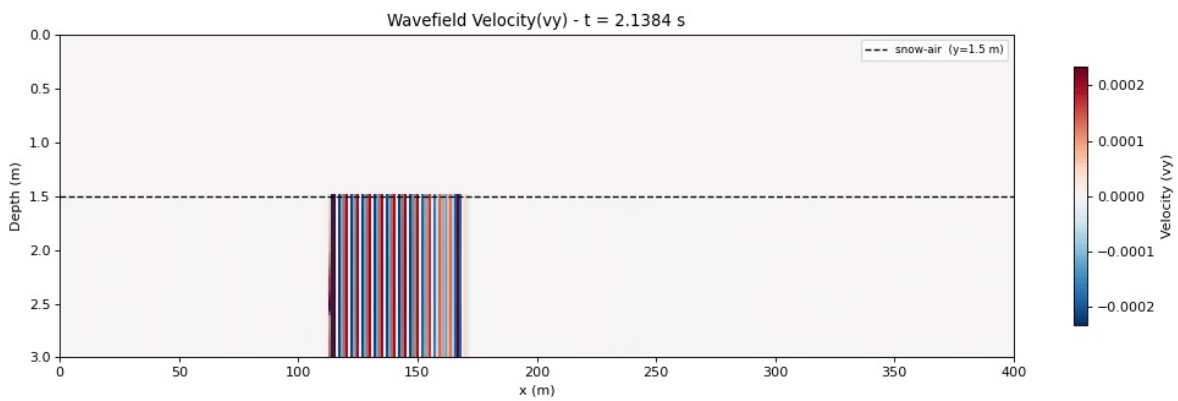


Figure F.2: Snapshot of the vertical wavefield velocity component (v_y) evaluated at the crack location ($y = 2.625$ m).

F.0.2 MPM-Guided Model

The stills of the animations of the velocity wavefield propagation for the MPM-guided simulations are shown below.

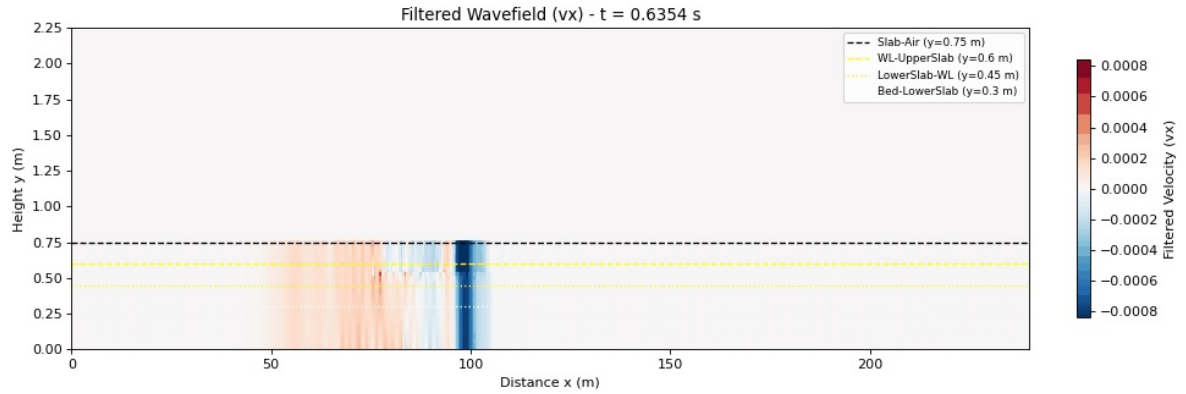


Figure F.3: Snapshot of the horizontal wavefield velocity component (v_x) for the MPM-guided model evaluated at the crack location ($y = 0.525$ m).

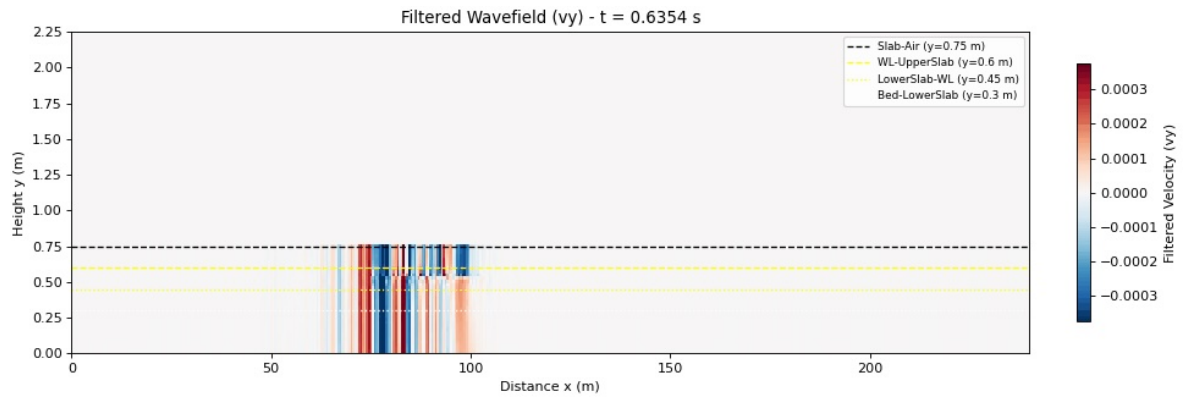


Figure F.4: Snapshot of the vertical wavefield velocity component (v_y) for the MPM-guided model evaluated at the crack location ($y = 0.525$ m).

The animations can also be found on the github repository in the folder Wave Propagation GIFS <https://github.com/s-bachmann/Master-Thesis/tree/main/Wave%20Propagation%20GIFS>.

Lamb wave propagation figure

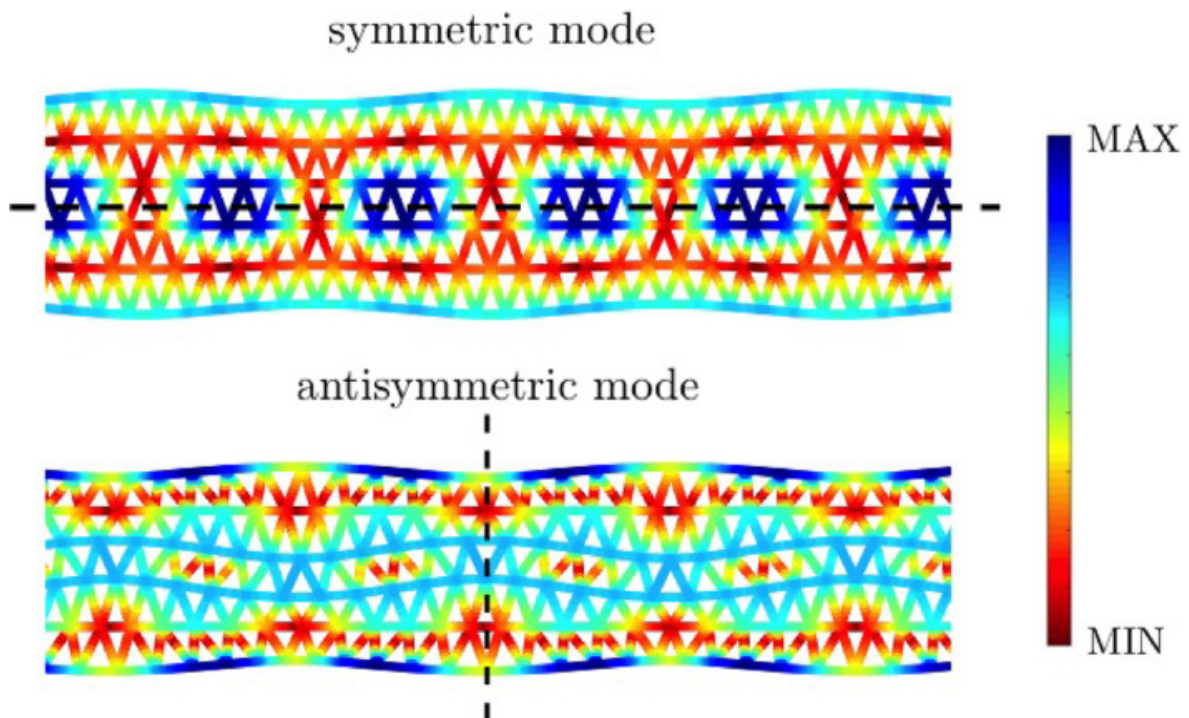


Figure G.1: Lamb wave propagation modes in plates. In the colour scale, blue (red) colour indicates larger (smaller) values of the total displacement field. Symmetric (S0) mode at top, antisymmetric (A0) mode at bottom. Figure taken from cartaLambWavesDiscrete2023.

Different DAS gauge length

The following figures show simulated DAS data of the simple model with a gauge length of 1m.

Sub-Rayleigh: Simulated DAS Strain Rate and Strain at $y = 2.625\text{m}$ and $y = 2.25\text{m}$

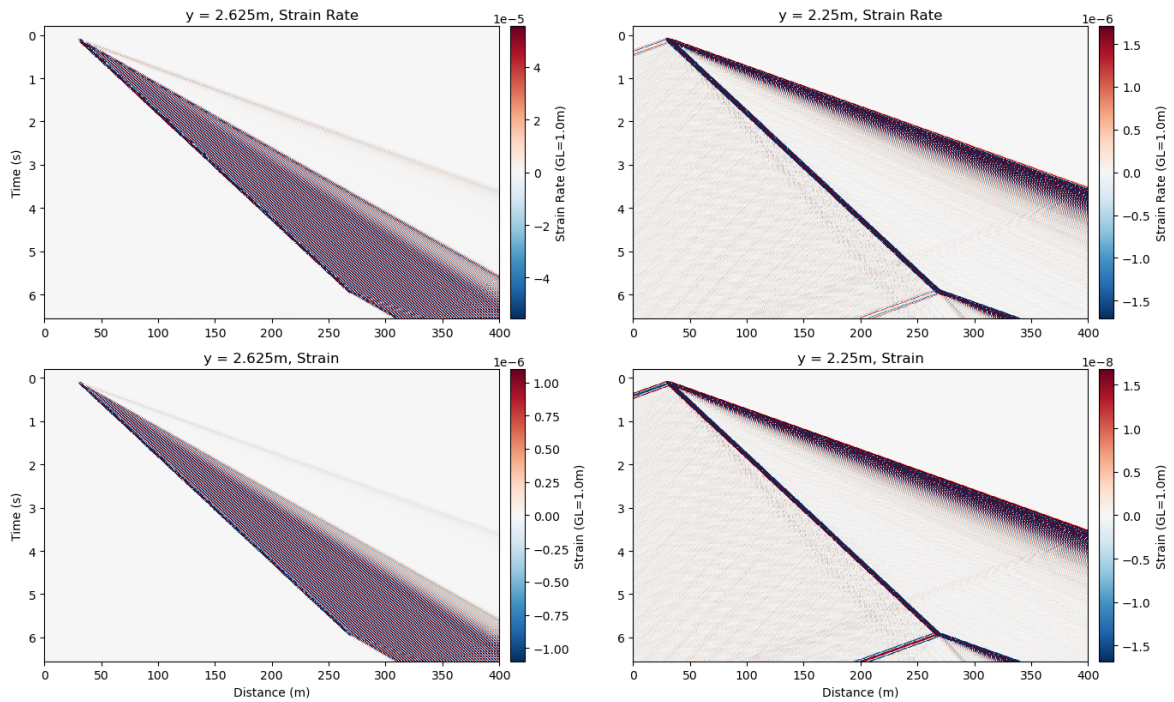


Figure H.1: Sub-rayleigh strain and strain rate (horizontal component) averaged over 1 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

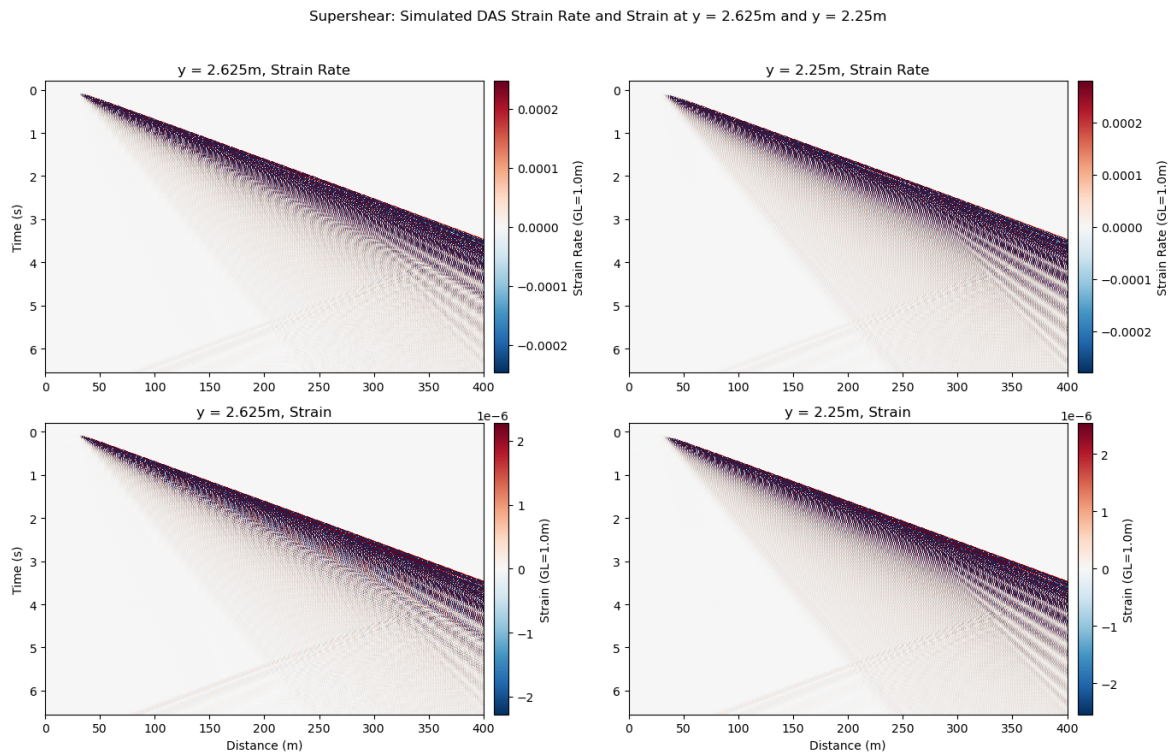


Figure H.2: Supershear strain and strain rate (horizontal component) averaged over 1 m gauge length to represent DAS data: strain rate in top column, strain in bottom column. Crack depth on left side, snow bottom on right side. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

MPM Displacement Plot

The following plot shows the displacement data extracted from the MPM simulation run. This data is shown to explain the noisyness of the velocity and strain data and to show that the output itself is not faulty.

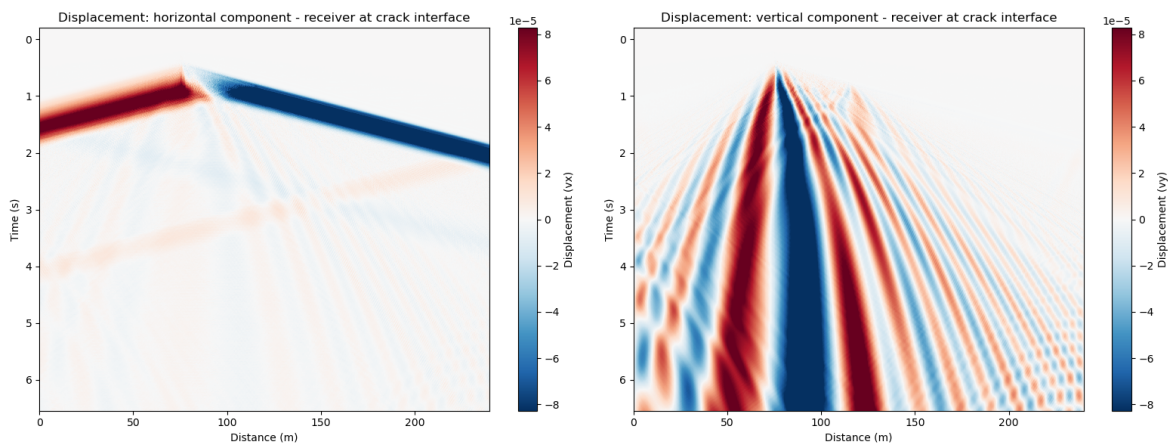


Figure I.1: Horizontal and vertical component displacement output from MPM-guided simulation, shown at crack interface. Colorbar shows magnitude of displacement.

Simple Source Setup in 5-Layer Model

The following figures show the simple sources (ricker wavelets with 0.1m spacing, sub-rayleigh case) in the same 5-layer model shown in figure ??.

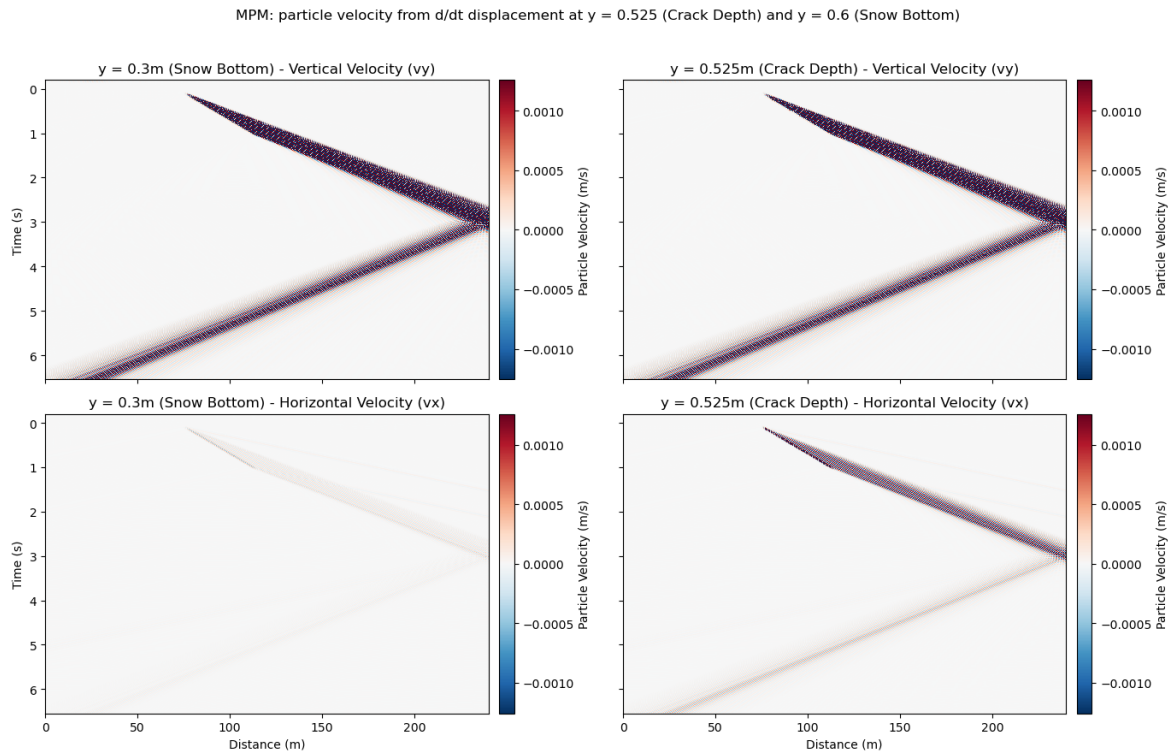


Figure J.1: Sub-rayleigh velocity. Crack depth in left column, snow bottom in right column. Model used is 5 layer model, source spacing is 0.1m and stf is a ricker wavelet. Colorbar shows velocity (m/s).

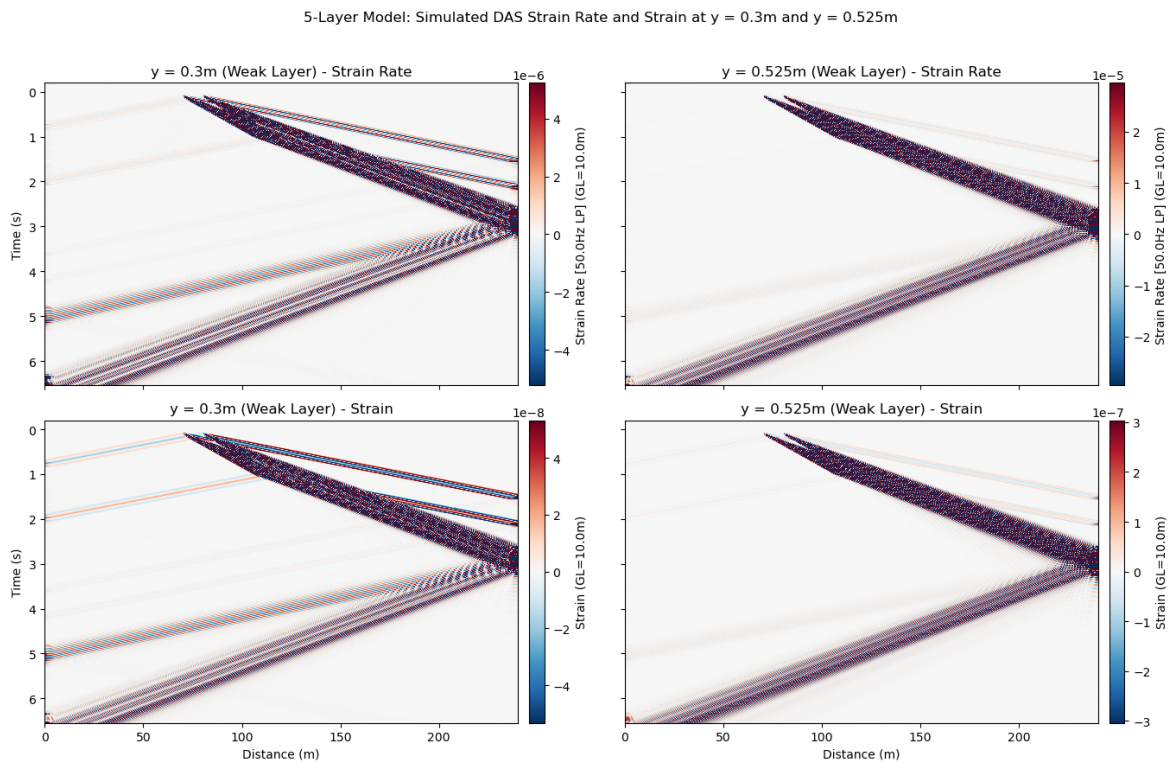


Figure J.2: Sub-rayleigh strain and strain rate. Crack depth in left column, snow bottom in right column. Model used is 5 layer model, source spacing is 0.1m and stf is a ricker wavelet. Colorbar shows magnitude of strain or strain rate, colorbars unscaled.

Temporary page!

$\LaTeX$ was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because $\LaTeX$ now knows how many pages to expect for this document.